

Gaussian processes & non-Gaussian likelihoods

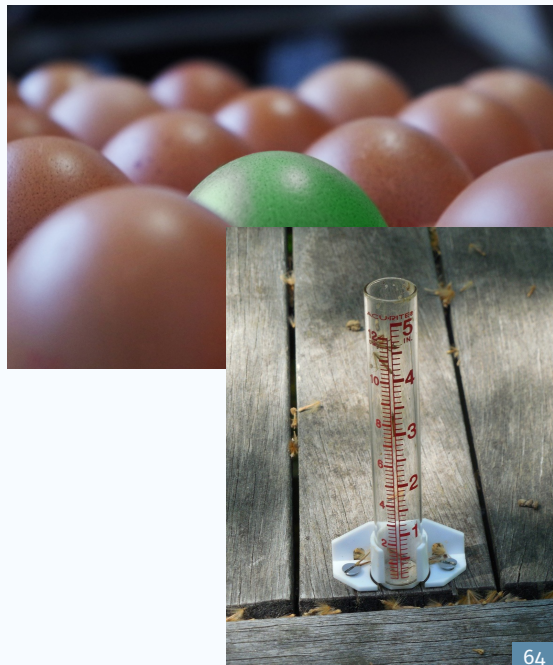
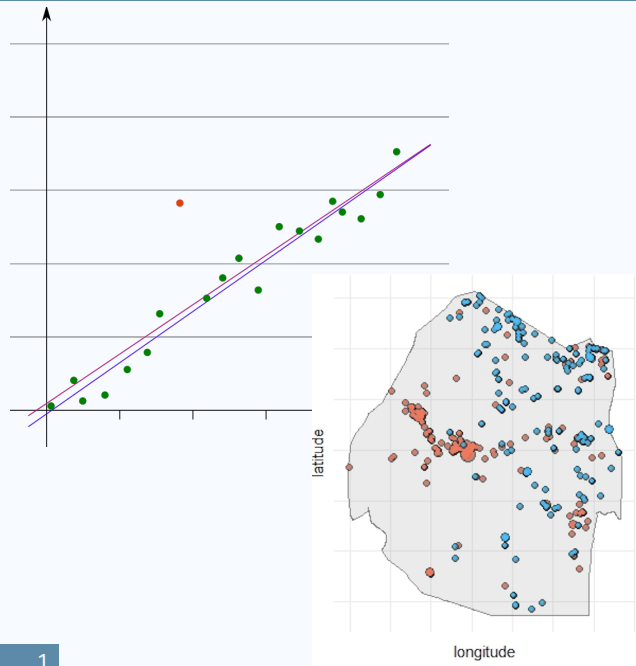
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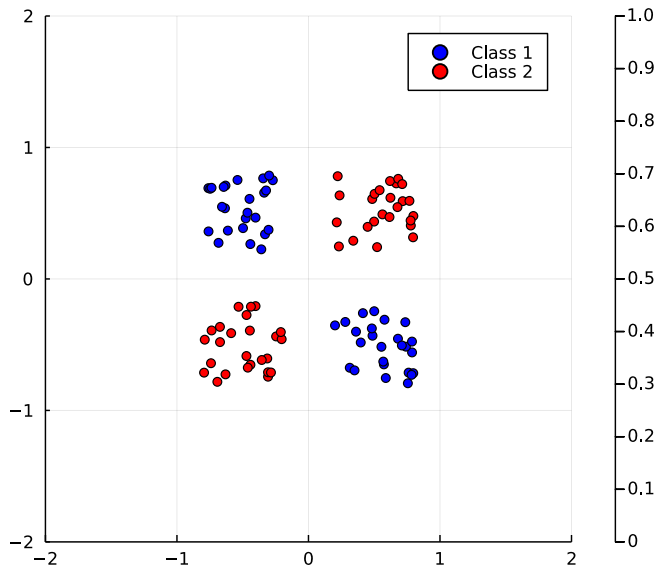
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Finnish Center for Artificial Intelligence
& Aalto University

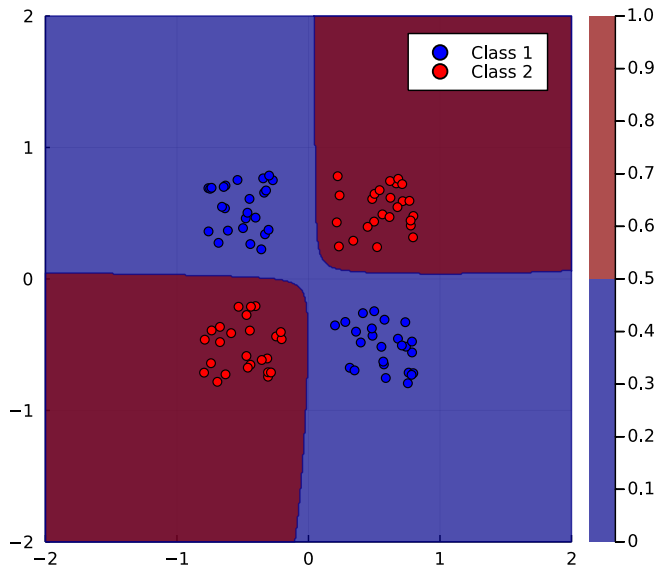
Gaussian Process Summer School 2024, 10 September 2024

Not Gaussian noise

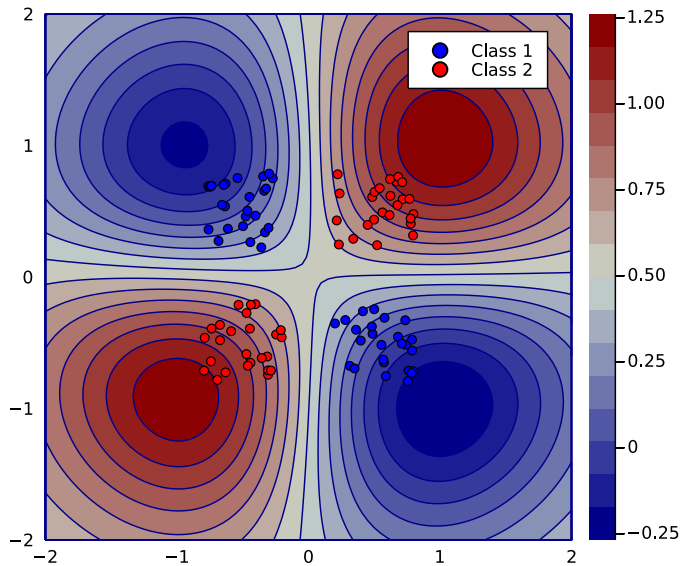




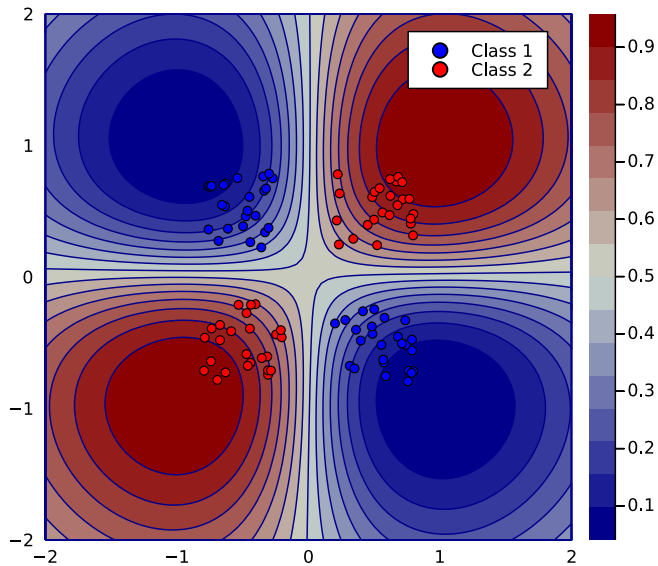
How can we model this?



SVM classification



Gaussian process regression

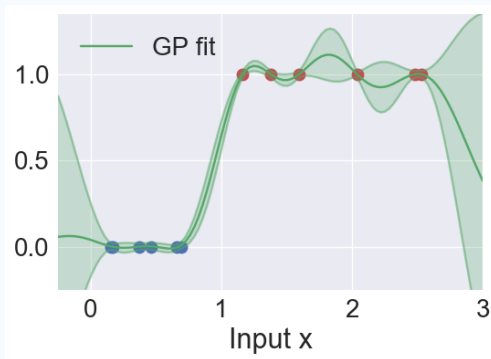


Gaussian process **classification**

Why don't we use regression models for classification?

- Binary classification: data set $\{x_n, y_n\}_{n=1}^N$ with $y_n \in \{0, 1\}$
- We want to model $p(y_n = +1 | x_n)$
- Why not simply use a GP regression model with labels: $y_n \in \{0, 1\}$:

$$p(y_n = +1 | x_n) = f(x_n)$$



Outline:

1. **Gaussian processes with Gaussian likelihood**
2. What is the likelihood? Connecting observations and Gaussian process prior
3. Non-Gaussian likelihoods: what happens to the posterior?
4. How to approximate the intractable
5. Comparison

- + *Intuitive* understanding
- + Learning the language

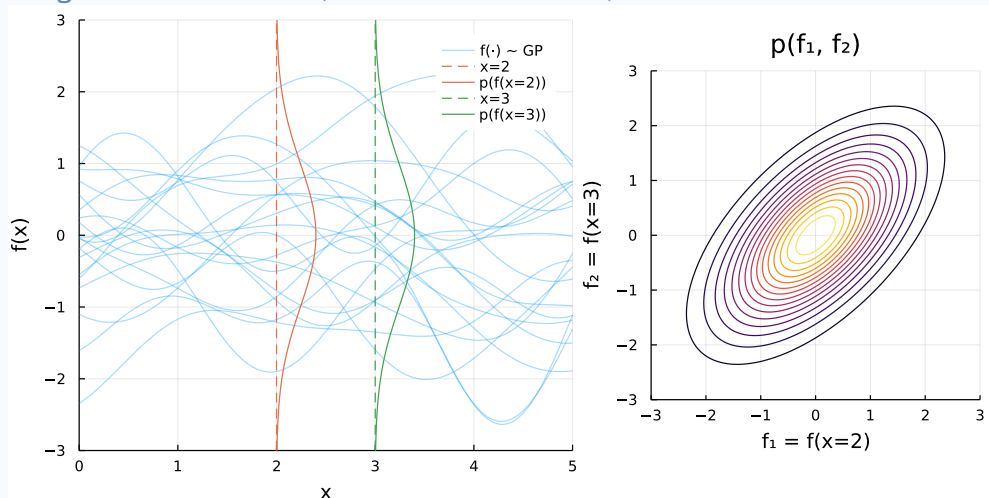
- In-depth expertise
- Lots of maths

Setting the scene

Gaussian process $f(\cdot)$

Distribution over functions

Marginals are Gaussian (mean and covariance)

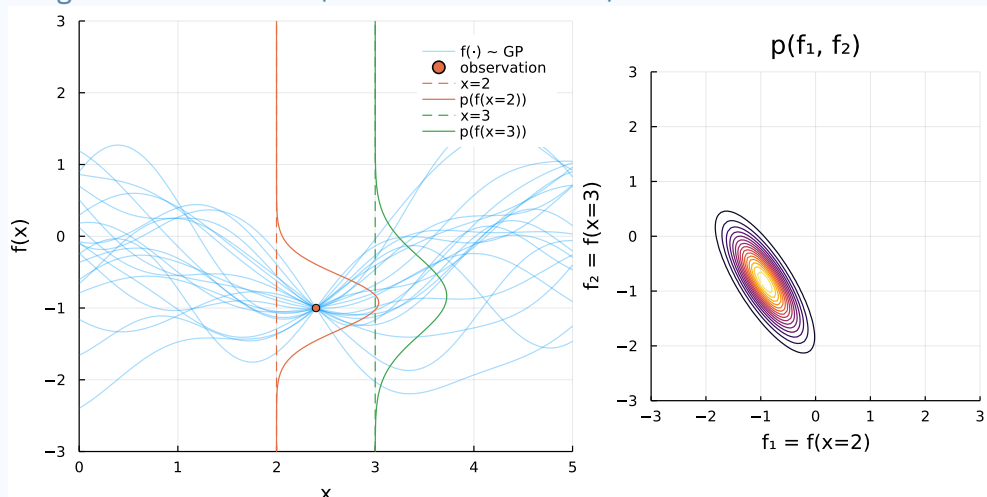


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Gaussian process conditioned on observation

Distribution over functions

Marginals are Gaussian (mean and covariance)

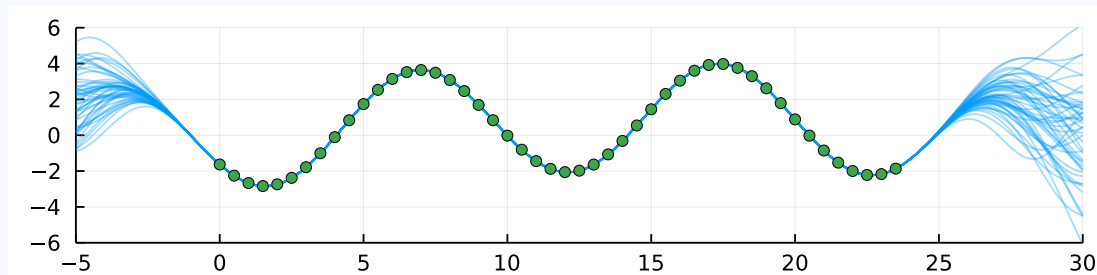


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exact conditioning

Without noise model, we interpolate observations:

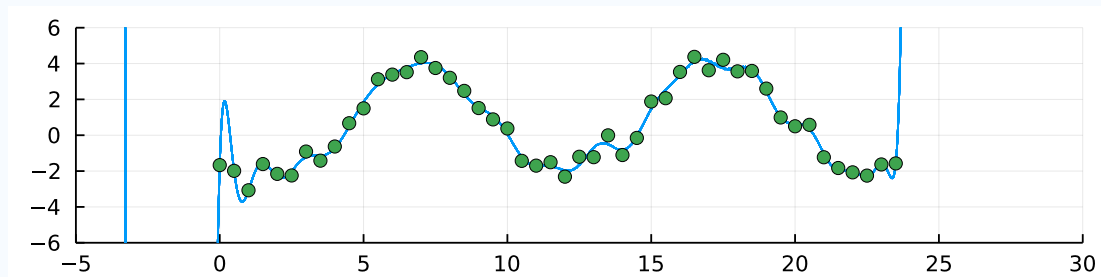
$$y(x) = f(x) + \epsilon, \quad \epsilon \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma_{\text{noise}}^2)$$
$$p(y|f) = \mathcal{N}(y|f, \sigma_{\text{noise}}^2)$$



exact conditioning

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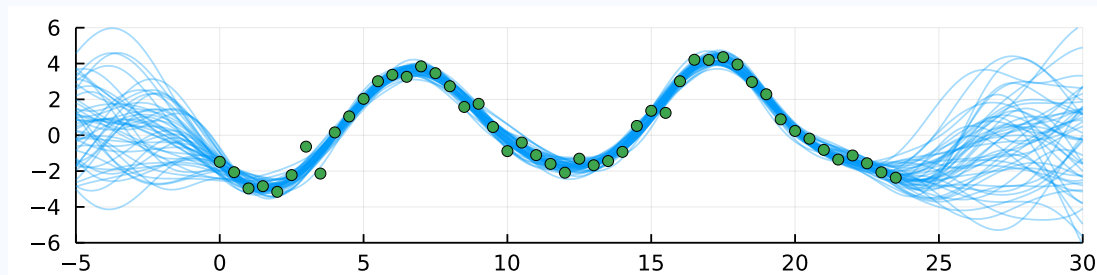
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Gaussian noise model

Gaussian additive noise model, written two ways:

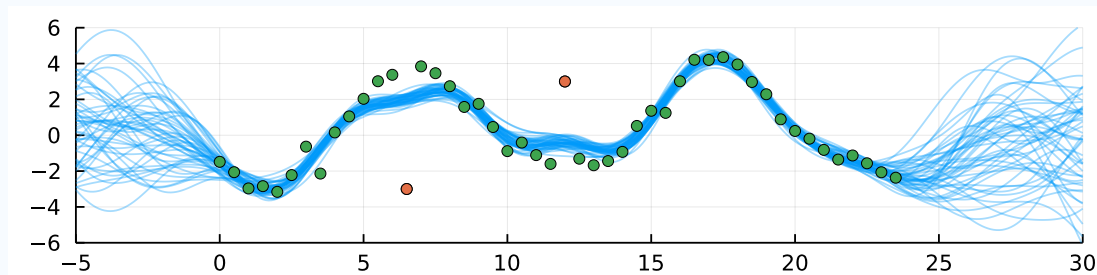
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misspecified Gaussian noise model

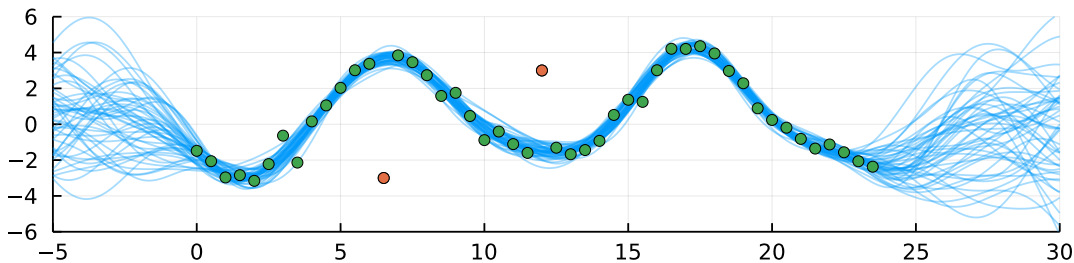
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heavy-tailed noise model

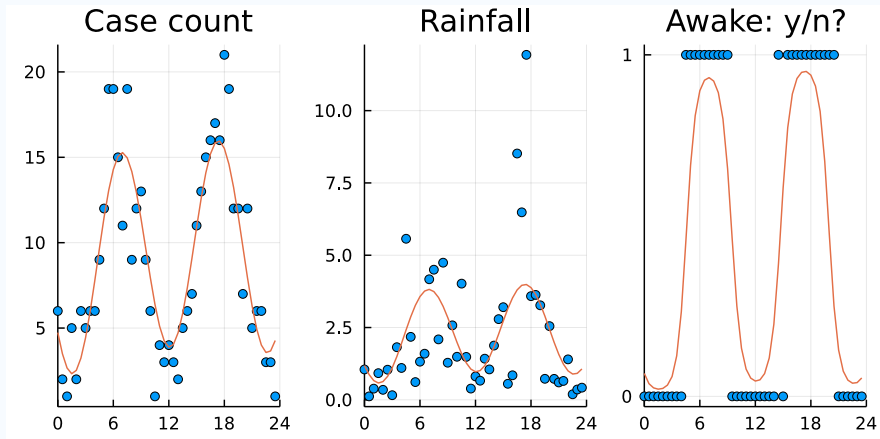
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- ✓ Gaussian processes with Gaussian likelihood
- 2. **What is the likelihood? Connecting observations and Gaussian process prior**
- 3. Non-Gaussian likelihoods: what happens to the posterior?
- 4. How to approximate the intractable
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Likelihood

Non-Gaussian observations



latent functional relationship (correlations!)
 $p(y_n | f(x_n))$

Likelihood

$$p(\mathbf{y} | \mathbf{f}) = \prod_{n=1}^N p(y_n | f_n); \quad f_n = f(x_n)$$

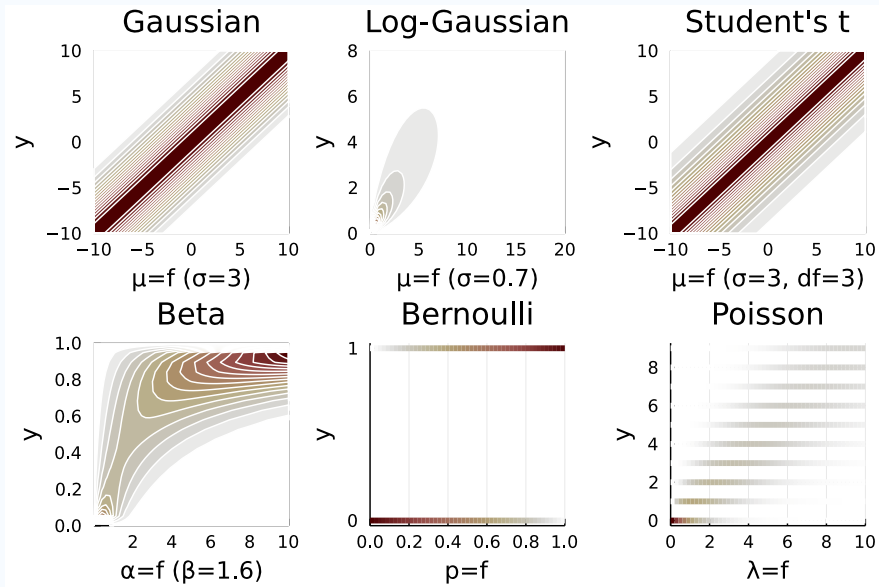
factorizing

Let's consider the individual (marginal, 1D) likelihood term:

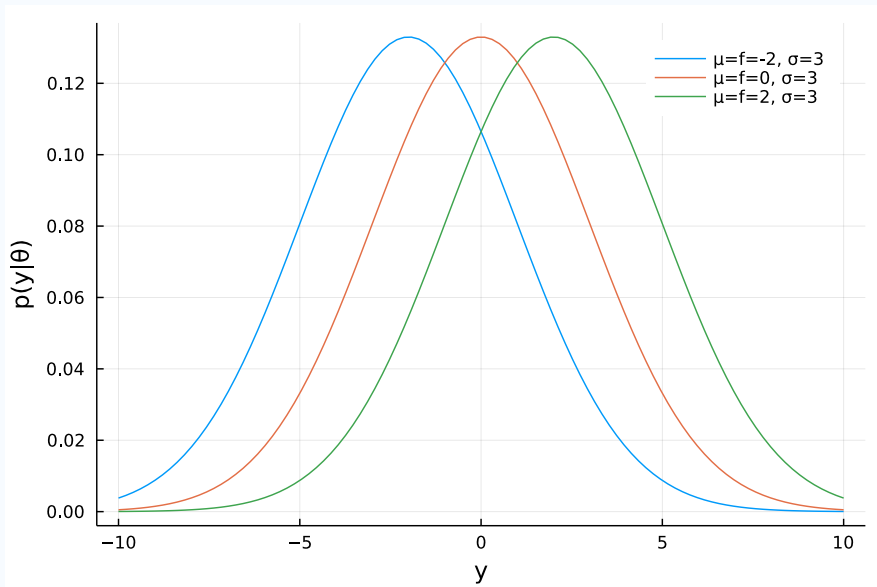
$$p(y | f)$$

Function of two arguments:

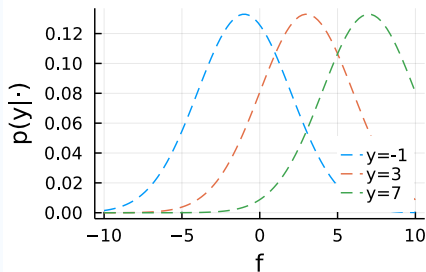
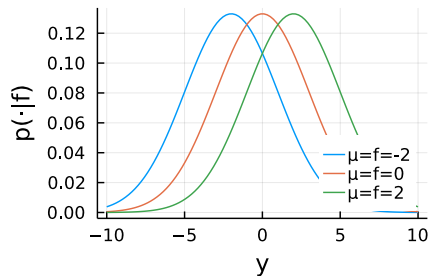
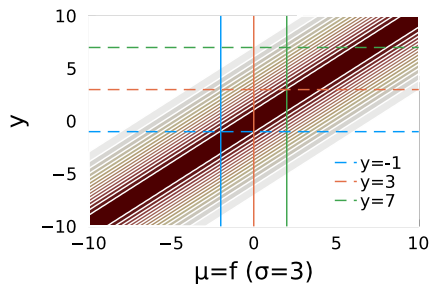
$$y \mapsto p(y | f), \quad f \mapsto p(y | f)$$



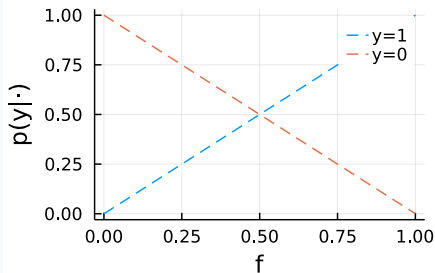
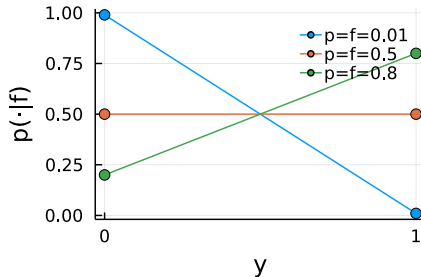
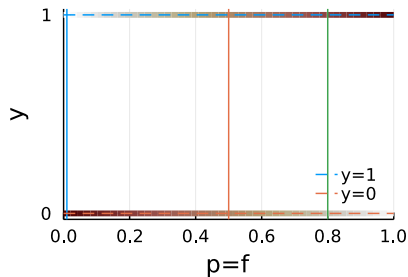
$p(y|f)$: Gaussian



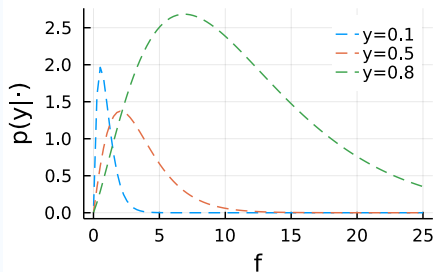
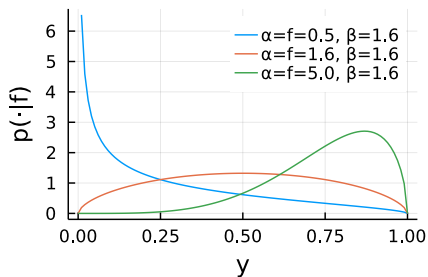
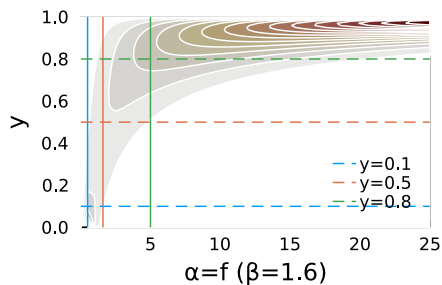
$p(y|f)$: Gaussian

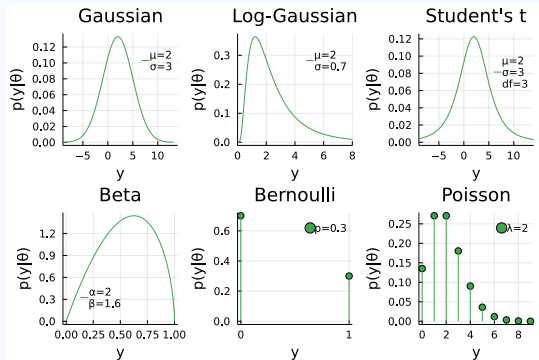
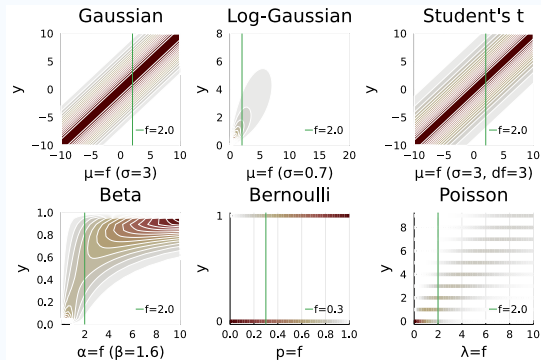


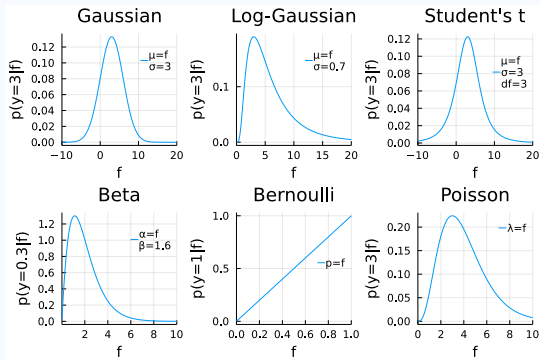
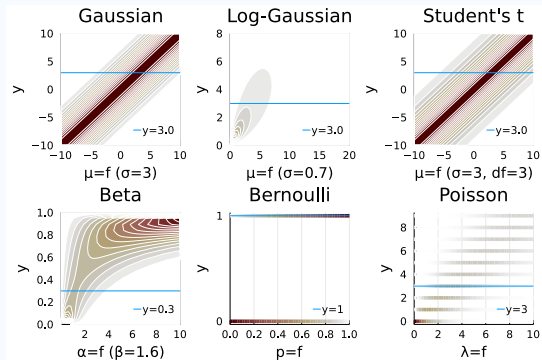
$p(y|f)$: Bernoulli



$p(y|f)$: Beta







Two important aspects of likelihoods:

1. link functions
2. log-concavity

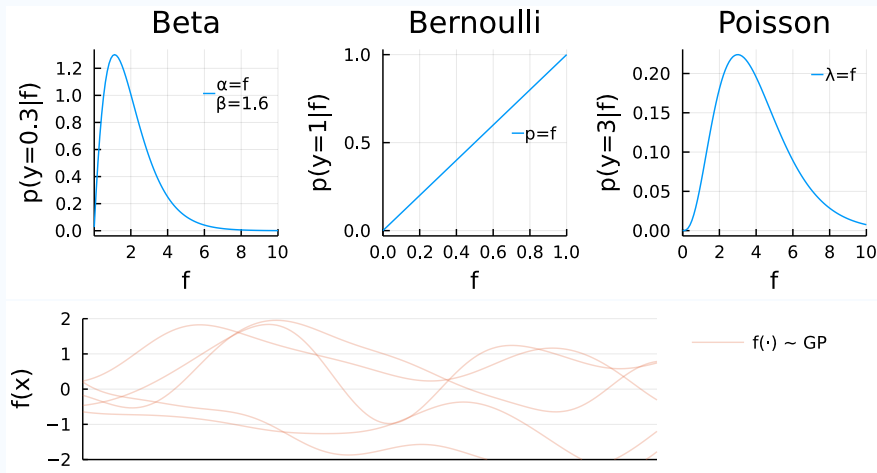
Link functions

$$\mathbb{E}[y] = \theta \in (0 \dots \infty)$$

$$f \sim \mathcal{N} \quad \in (-\infty \dots \infty)$$

$$\text{link}(\theta) = f$$

$$\theta = \text{invlink}(f)$$



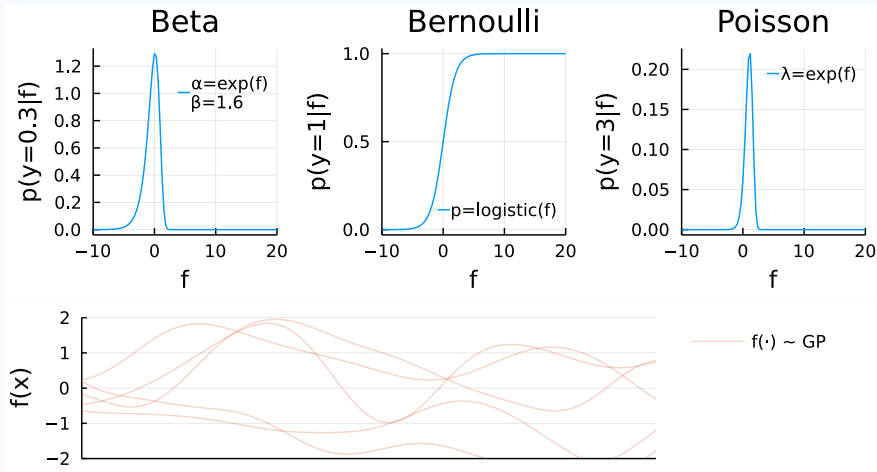
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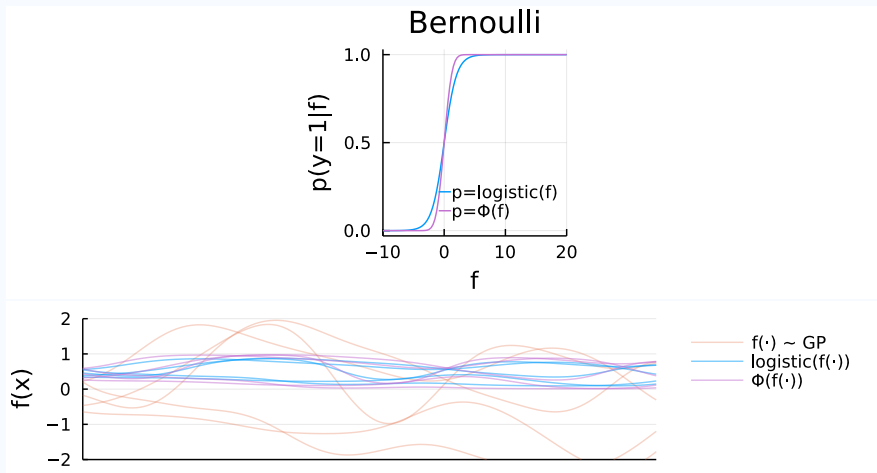
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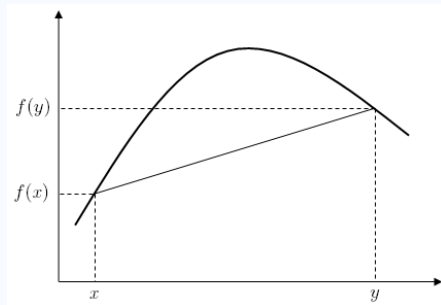
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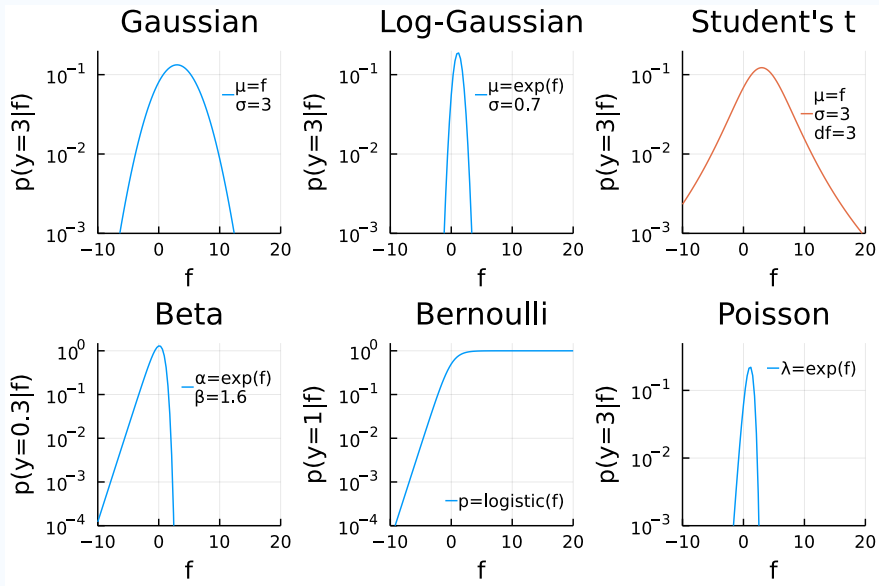


(Log-)concavity

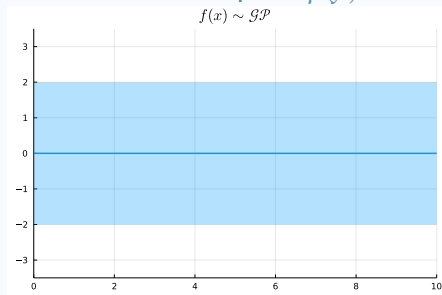


$$f(\alpha x + (1 - \alpha)y) \geq \alpha f(x) + (1 - \alpha)f(y)$$

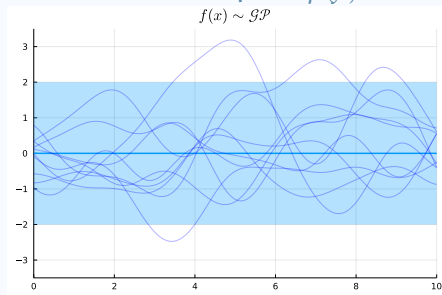
Log-concavity of likelihoods



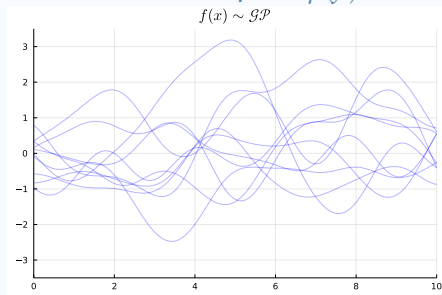
Functional prior $p(f)$



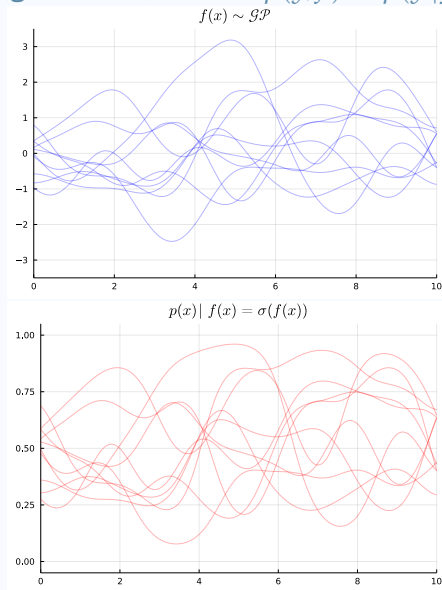
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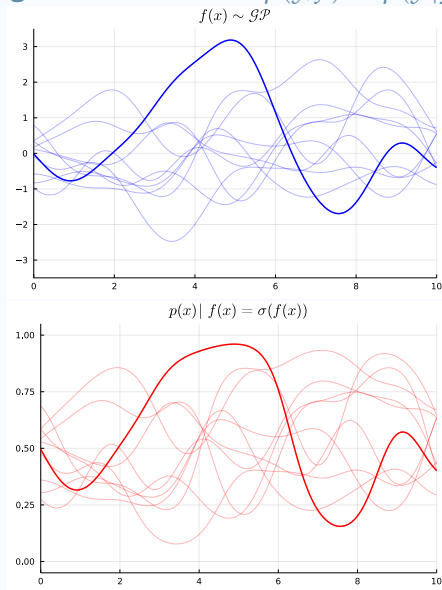
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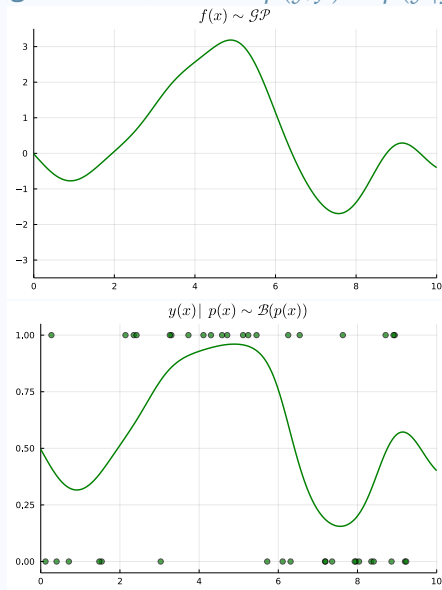
Joint (generative) model: $p(y, f) = p(y | f)p(f)$



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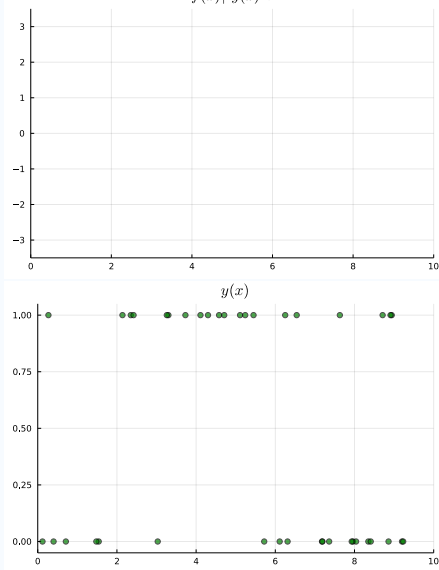


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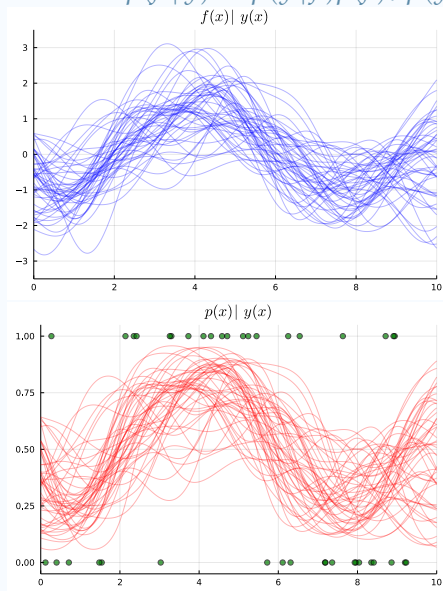
Back to GPs...

Posterior: $p(f | y) = p(y | f)p(f) / p(y)$
 $f(x) | y(x) ?$

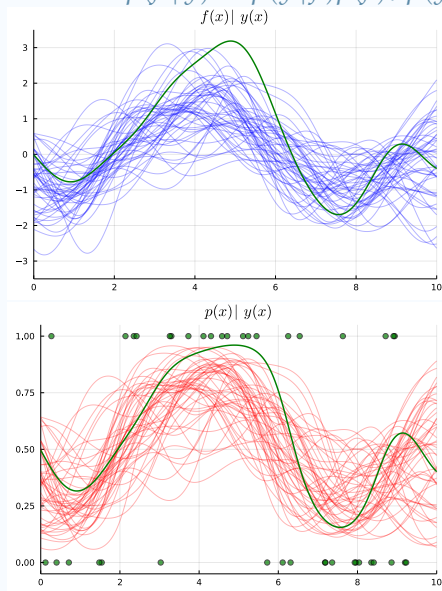


Back to GPs...

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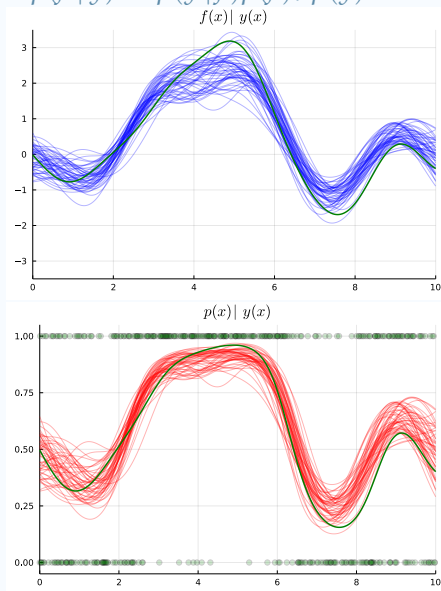


Posterior: $p(f | y) = p(y | f)p(f) / p(y)$



Back to GPs...

Posterior: $p(f | y) = p(y | f)p(f) / p(y)$ for more data



- ✓ Gaussian processes with Gaussian likelihood
- ✓ What is the likelihood? Connecting observations and Gaussian process prior
- 3. **Non-Gaussian likelihoods: what happens to the posterior?**
- 4. How to approximate the intractable
- 5. Comparison

Posterior

Likelihood

$$p(\mathbf{y} | \mathbf{f})$$

Joint distribution

$$p(\mathbf{y}, \mathbf{f}) = p(\mathbf{y} | \mathbf{f})p(\mathbf{f})$$

Posterior

$$\mathbf{f} \mapsto p(\mathbf{f} | \mathbf{y}) = \frac{p(\mathbf{y} | \mathbf{f})p(\mathbf{f})}{p(\mathbf{y})}$$

$$\mathbf{y} \mapsto (\mathbf{f} \mapsto p(\mathbf{f} | \mathbf{y}))$$

Posterior predictions

At new point x^* :

$$p(f^* | x^*, \mathbf{x}, \mathbf{y}) = \int p(f^* | x^*, \mathbf{x}, \mathbf{f}) p(\mathbf{f} | \mathbf{x}, \mathbf{y}) d\mathbf{f}$$

At training data:

$$p(\mathbf{f} | \mathbf{x}, \mathbf{y}) = \frac{p(\mathbf{f} | \mathbf{x}) \prod_{n=1}^N p(y_n | f(x_n))}{\int p(\mathbf{f}' | \mathbf{x}) \prod_{n=1}^N p(y_n | f'(x_n)) d\mathbf{f}'}$$

$$p(\mathbf{f} | \mathbf{y}) = \frac{1}{Z} p(\mathbf{f}) \prod_{n=1}^N p(y_n | f_n)$$

$$Z = p(\mathbf{y} | \mathcal{M}) = \int p(\mathbf{f} | \mathcal{M}) \prod_{n=1}^N p(y_n | f_n, \mathcal{M}) d\mathbf{f}$$

“marginal likelihood” or “evidence” given **model** $\mathcal{M}$

$$p(\mathbf{f} | \mathbf{y}) = \frac{1}{Z} p(\mathbf{f}) \prod_{n=1}^N p(y_n | f_n)$$

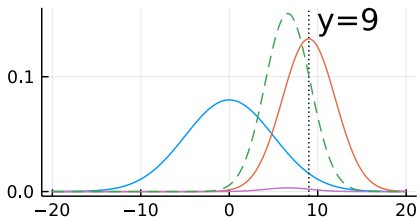
Gaussian (process) prior $p(f(\cdot)) \dots$ $p(\mathbf{f}) = \mathcal{N}(\mathbf{f} | \mathbf{0}, \mathbf{K})$

& Gaussian likelihood: conjugate case $\rightarrow$ posterior Gaussian

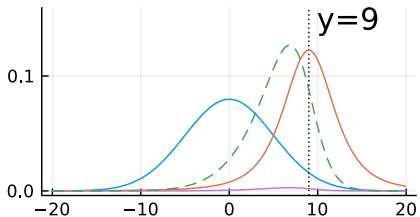
& **non**-Gaussian $p(y|f)$ $\rightarrow p(\mathbf{f} | \mathbf{y})$ also **non**-Gaussian, **intractable**

1D examples

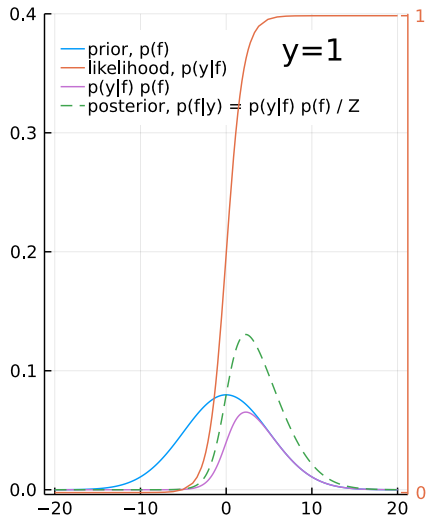
Gaussian



Student's t

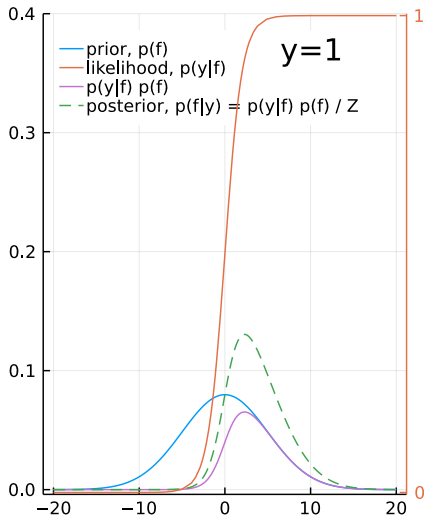


Bernoulli

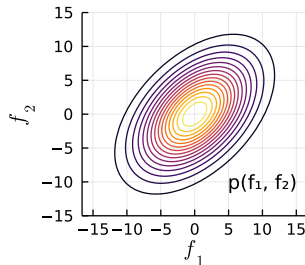


Bernoulli example in 2D

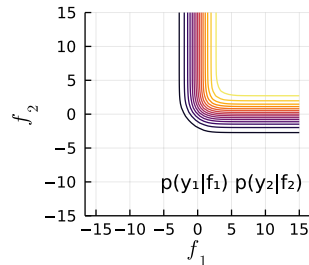
Bernoulli



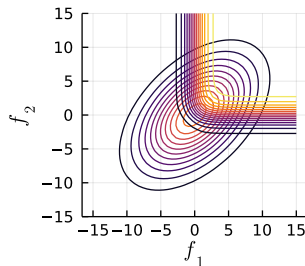
Prior



Likelihood



Posterior



Posterior for N observations

$$p(\mathbf{f} | \mathbf{y}) = \frac{p(\mathbf{f}) \prod_{n=1}^N p(y_n | f_n)}{\int p(\mathbf{f}') \prod_{n=1}^N p(y_n | f'_n) d\mathbf{f}'}$$

$$f_1 = f(x_1)$$

$$f_2 = f(x_2)$$

$$\vdots$$

$$f_N = f(x_N)$$

Summary so far

- What is the likelihood $p(y|f)$?
- When is it non-Gaussian?
- Why does the posterior $p(f|y)$ become intractable?

Questions?! :)

- ✓ Gaussian processes with Gaussian likelihood
- ✓ What is the likelihood? Connecting observations and Gaussian process prior
- ✓ Non-Gaussian likelihoods: what happens to the posterior?
- 4. **How to approximate the intractable**
- 5. Comparison

Approximations

■ Joint model:

$$p(\mathbf{y}, \mathbf{f}) = p(\mathbf{y} | \mathbf{f}) p(\mathbf{f}) = \prod_{n=1}^N p(y_n | f_n) \mathcal{N}(\mathbf{f} | \mathbf{0}, \mathbf{K})$$

■ Posterior distribution at training points:

$$p(\mathbf{f} | \mathbf{y}) = \frac{p(\mathbf{y} | \mathbf{f}) p(\mathbf{f})}{p(\mathbf{y})} \approx q(\mathbf{f})$$

■ Posterior of f^* for new test point $\mathbf{x}^*$:

$$p(f^* | \mathbf{y}) = \int p(f^* | \mathbf{f}) p(\mathbf{f} | \mathbf{y}) d\mathbf{f} \approx \int p(f^* | \mathbf{f}) q(\mathbf{f}) d\mathbf{f} \equiv q(f^*)$$

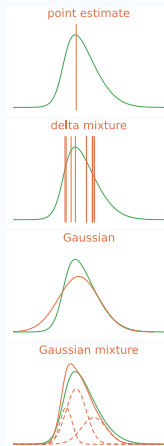
■ Predictive distribution

$$p(y^* | \mathbf{y}) = \int p(y^* | f^*) p(f^* | \mathbf{y}) df^* \approx \int p(y^* | f^*) q(f^*) df^*$$

Analytically intractable distributions!

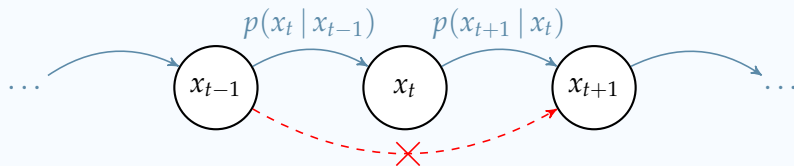
Approximating distributions

- Delta distribution
 - ▶ Point estimate
- Mixture of delta distributions
 - ▶ **Markov Chain Monte Carlo (MCMC)**
 - ▶ Neural network ensembles...
- Gaussian distribution
 - ▶ Laplace Approximation (LA)
 - ▶ Variational Bayes/Variational Inference (VB / VI)
 - ▶ Expectation Propagation (EP), PowerEP, ...
- Mixture of Gaussians
- ...



Markov Chain Monte Carlo

Markov Chain



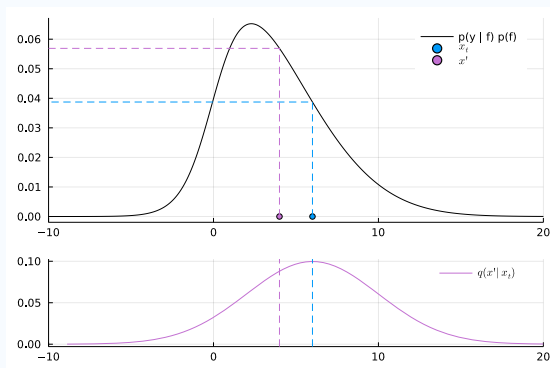
- Samples $x_1, \dots, x_T$
- “Markov” = 1-step history
- $x_{t+1} \sim p(x_{t+1} | x_t)$, independent of $x_{t-1}, \dots, x_1$

Markov Chain Monte Carlo (MCMC)

Generate samples $\{x_t\} \sim p(f | y)$

Requires:

- *unnormalized* posterior $h(f) = p(y|f)p(f)$
- Markov proposal $q(x' | x_t)$
- initial x_0



In each iteration t :

1. Random proposal $x' \sim q(x' | x_t)$
2. Acceptance probability $\frac{h(x')}{h(x_t)} \rightarrow$ ensures sampling from $p(f | y)$

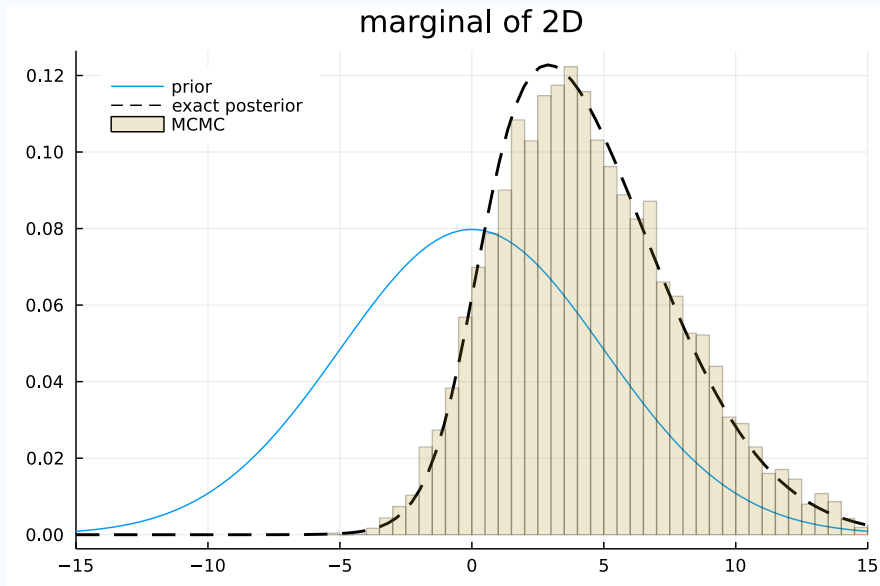
accept: $x_{t+1} = x'$ reject: copy $x_{t+1} = x_t$

$h(x') > h(x_t)$: always accepts $\rightarrow$ climbs uphill

Demo: MCMC in 2D

tinyurl.com/nongaussian-inference-viz-v1

Marginals



MCMC: important properties

- burn-in
- acceptance ratio
- auto-correlation, effective sample size (ESS); thinning to save memory
- mixing and multiple chains ($\hat{R}$)
- better proposals (HMC, NUTS) → use robust implementations
 - + very accurate (gold-standard)
 - very slow, predictions require keeping all (thinned) samples around

Michael Betancourt's betanalpha.github.io/writing/

MCMC: robust implementations

■ Stan



■ PyMC3



PyMC3

■ Pyro & NumPyro

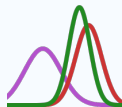


■ TensorFlow Probability (GPflow)



GPflow

■ Turing.jl



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- ✓ What is the likelihood? Connecting observations and Gaussian process prior
- ✓ Non-Gaussian likelihoods: what happens to the posterior?

4. **How to approximate the intractable**

- ✓ with samples: MCMC

4.2 **with Gaussians**

- Laplace Approximation
- Variational Inference
- Expectation Propagation

5. Comparison

Gaussian approximations

Approximating the exact posterior with Gaussian

Approximating the posterior at observations:

$$p(\mathbf{f} | \mathbf{y}) \approx q(\mathbf{f}) = \mathcal{N}(\mathbf{f} | \mu = ?, \Sigma = ?)$$

Predictions at new points:

$$p(f^* | x^*, \mathbf{y}) \approx q(f^*) = \underbrace{\int p(f^* | x^*, \mathbf{f}) q(\mathbf{f}) d\mathbf{f}}_{\text{closed-form integral!}}$$

Demo: What does this mean for Gaussian processes?

tinyurl.com/nongaussian-inference-viz-v1

Choosing μ and Σ for $q(\mathbf{f})$

$$p(\mathbf{f} | \mathbf{y}) \approx q(\mathbf{f}) = \mathcal{N}(\mathbf{f} | \mu = ?, \Sigma = ?)$$

locally: match mean &
variance at point

globally: minimise divergence

**Laplace
approximation**

Variational
Inference (VI)

Expectation
Propagation (EP)

Laplace approximation

Laplace approximation

Idea: log of Gaussian pdf = quadratic polynomial

$$p_{\mathcal{N}}(\mathbf{f}) = \frac{1}{\sqrt{(2\pi)^k |\Sigma|}} \exp\left(-\frac{1}{2}(\mathbf{f} - \mu)^{\top} \Sigma^{-1} (\mathbf{f} - \mu)\right)$$

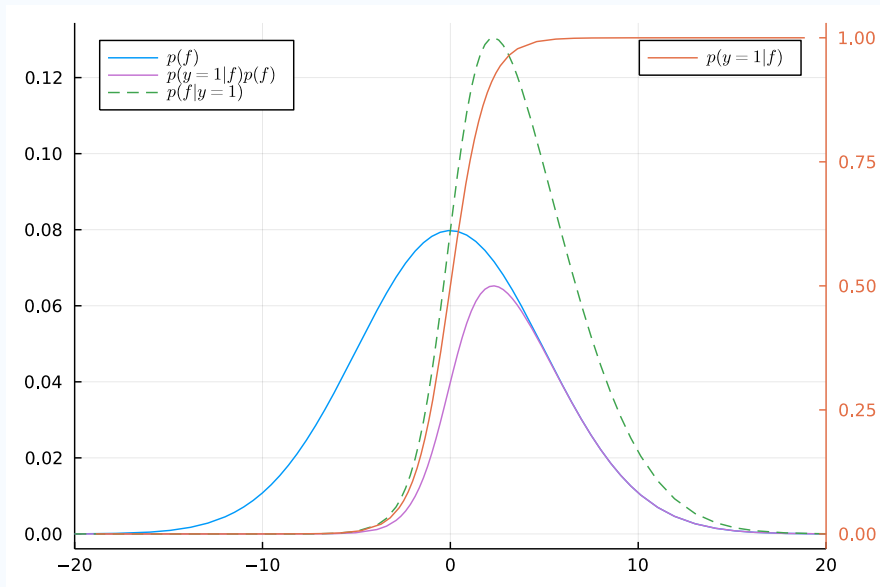
quadratic polynomial through approximation:

2nd-order Taylor expansion of log of $h(f) = p(y|f)p(f)$ at $\hat{f}$

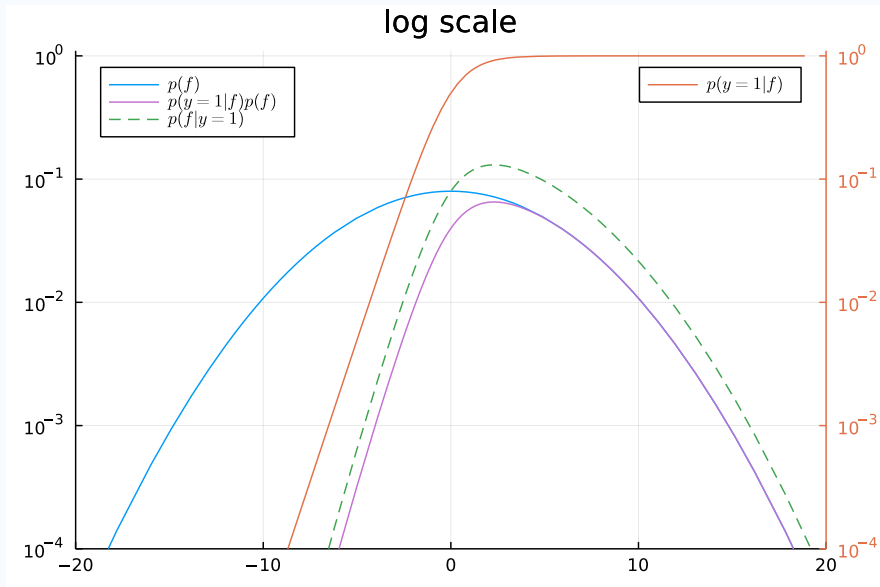
$$g(x + \delta) \approx g(x) + \left(\frac{dg}{dx}(x)\right)\delta + \frac{1}{2!}\left(\frac{d^2g}{dx^2}(x)\right)\delta^2$$

1. Find **mode** of posterior
2nd-order gradient optimisation (e.g. Newton's method)
2. Match **curvature** (Hessian) at mode

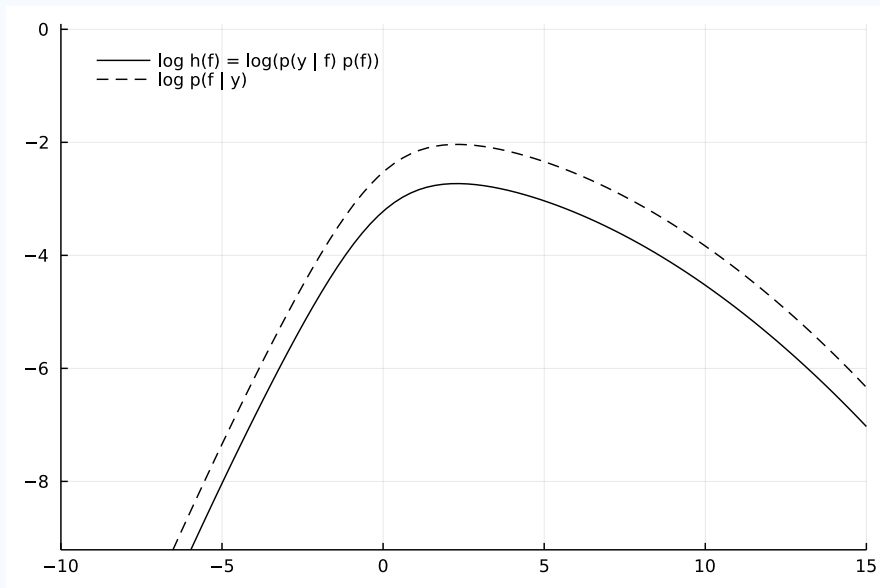
$$p(f | y) = \frac{1}{Z} p(y | f) p(f)$$



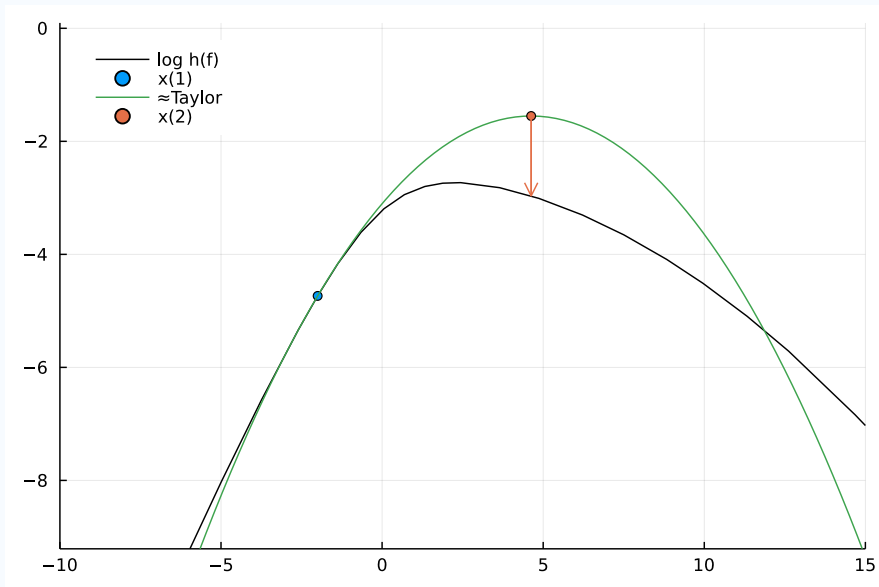
$$\log p(f | y) = -\log Z + \log p(y | f) + \log p(f)$$



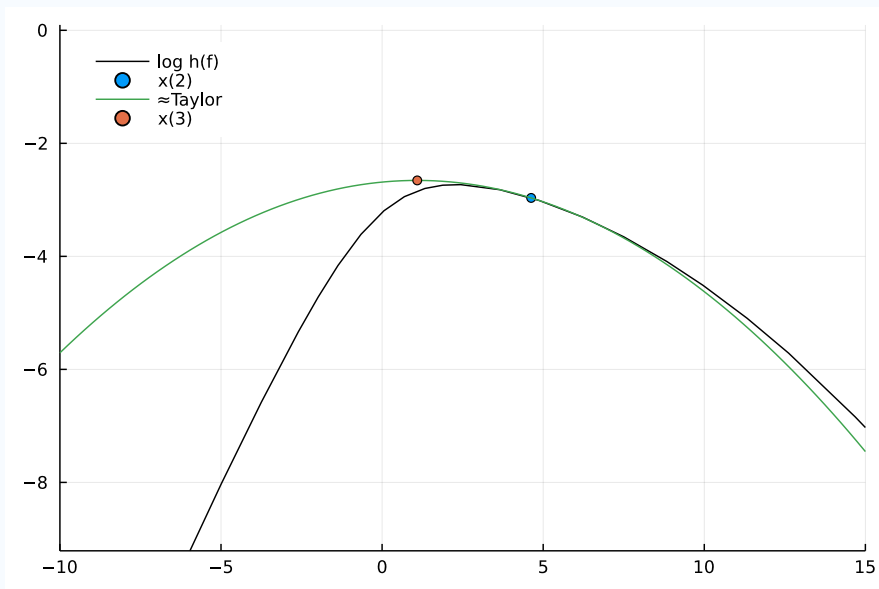
$$\log p(f | y) = -\log Z + \log h(f)$$



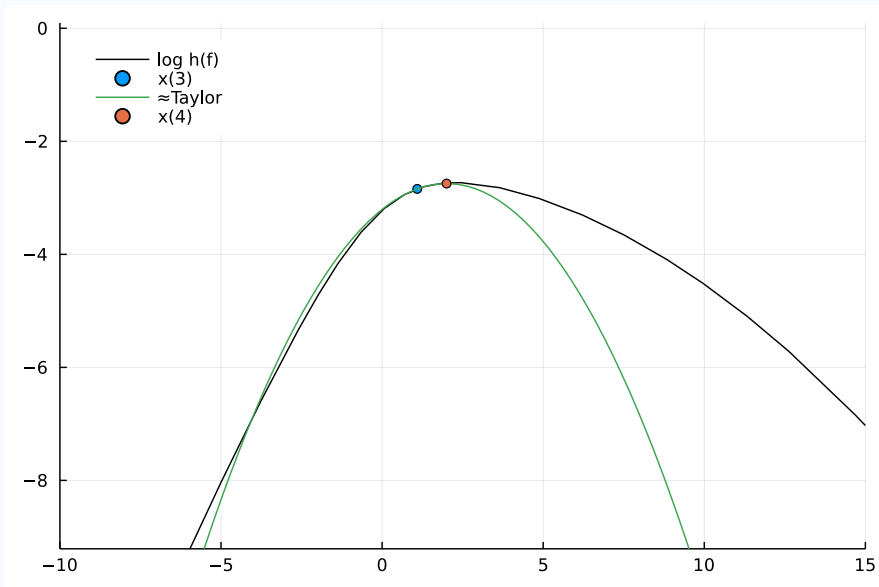
Newton's method



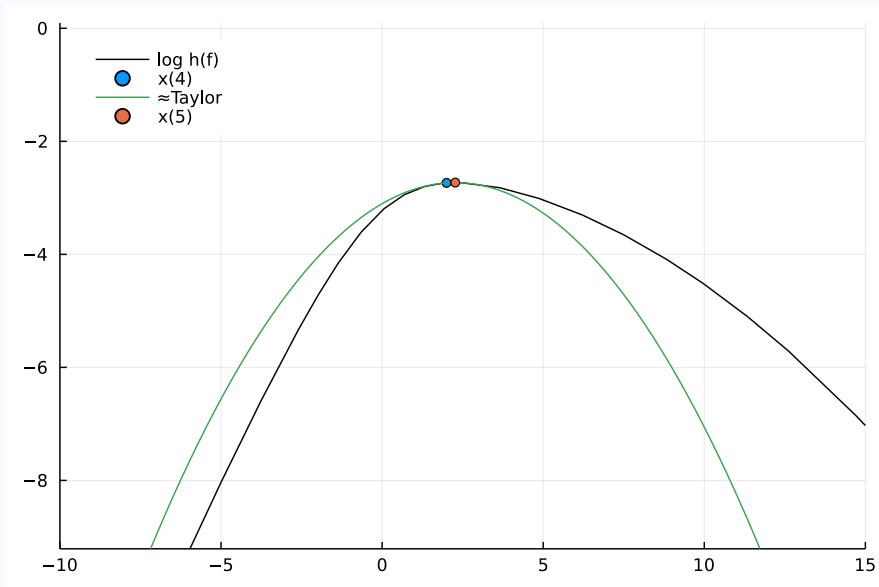
Newton's method



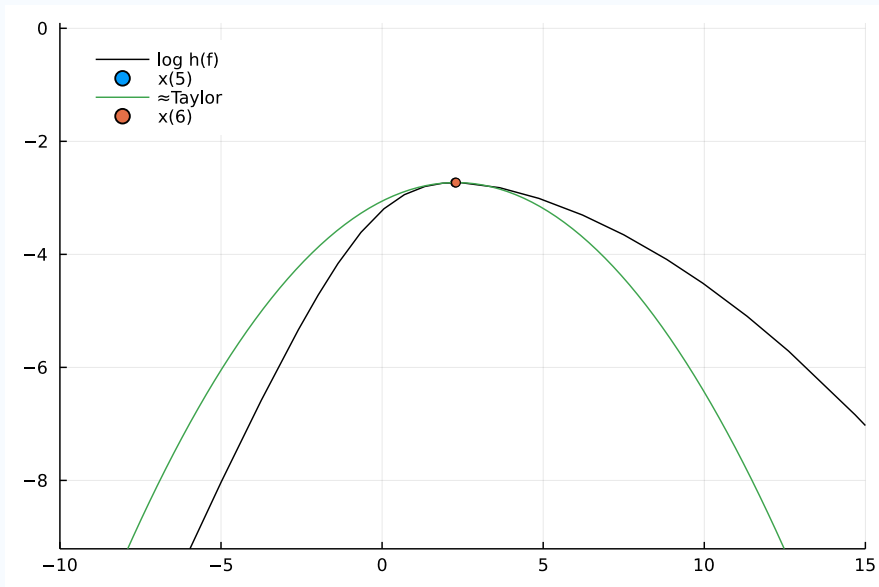
Newton's method



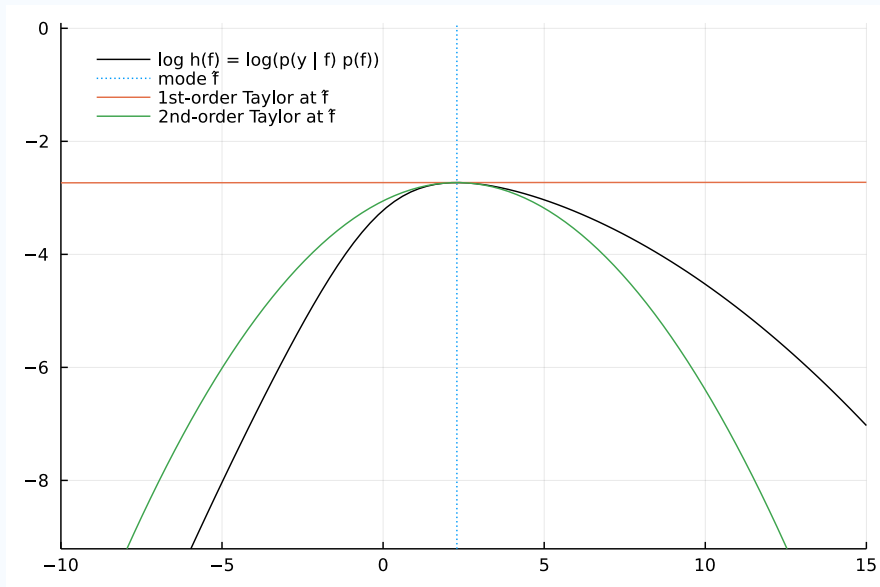
Newton's method



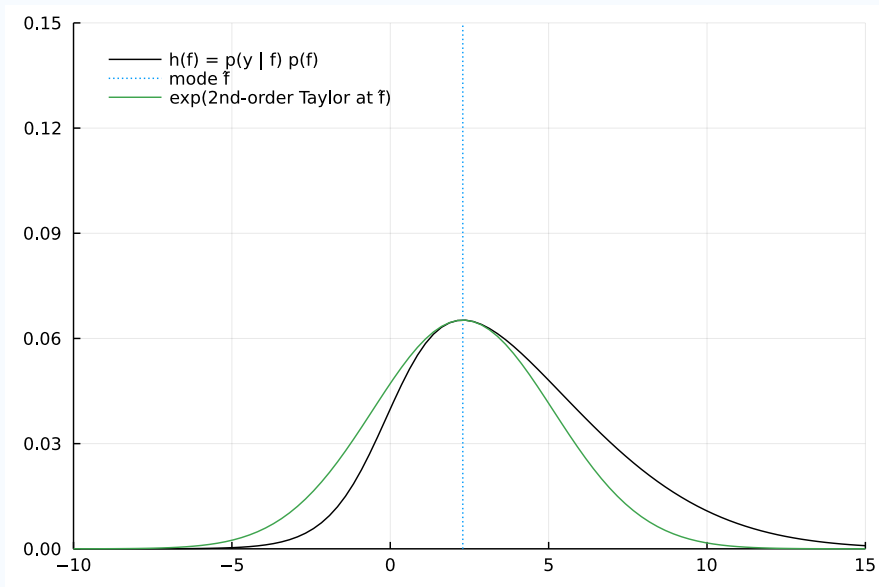
Newton's method



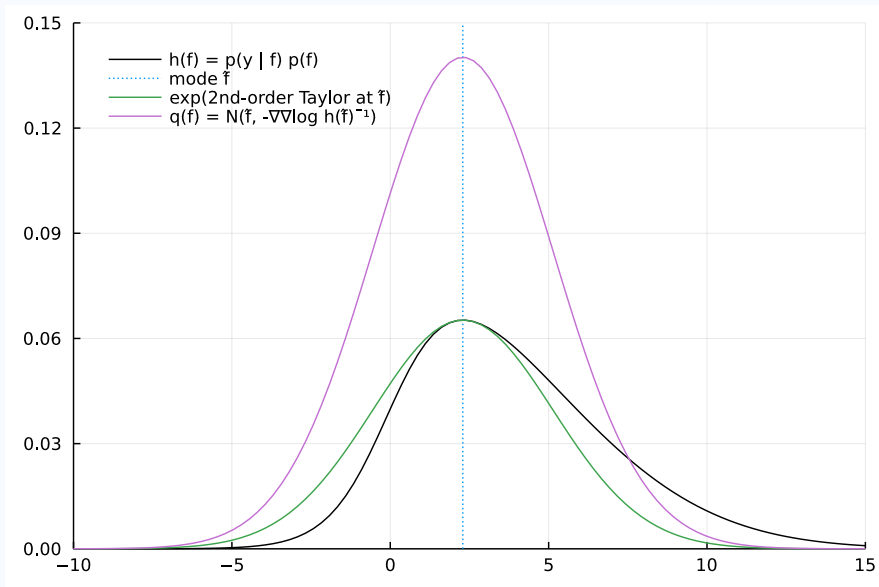
$$\log p(f | y) + \log Z = \log h(f) \approx \mathcal{O}(f^2)$$



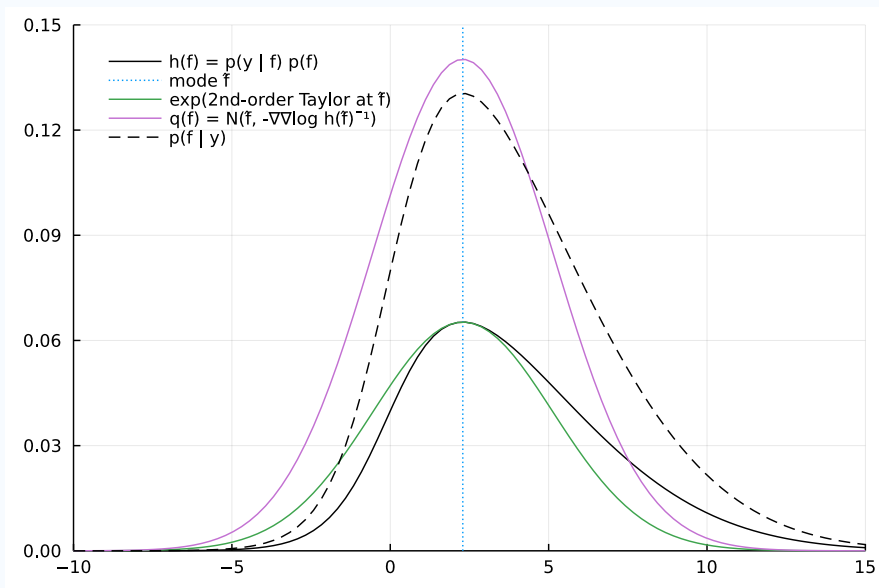
$$p(f | y) Z \approx \exp(\mathcal{O}(f^2))$$



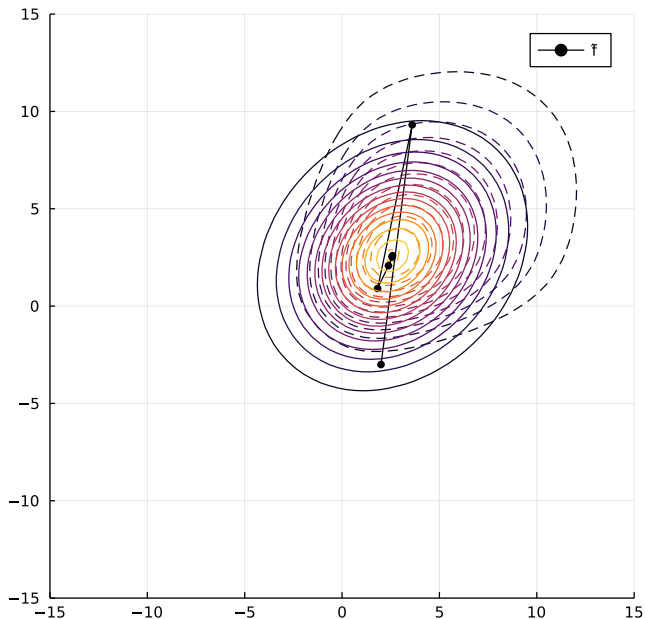
$$p(f | y) \approx \mathcal{N}(f | \hat{f}, -(\mathrm{d}^2 \log h / \mathrm{d}f^2)^{-1})$$



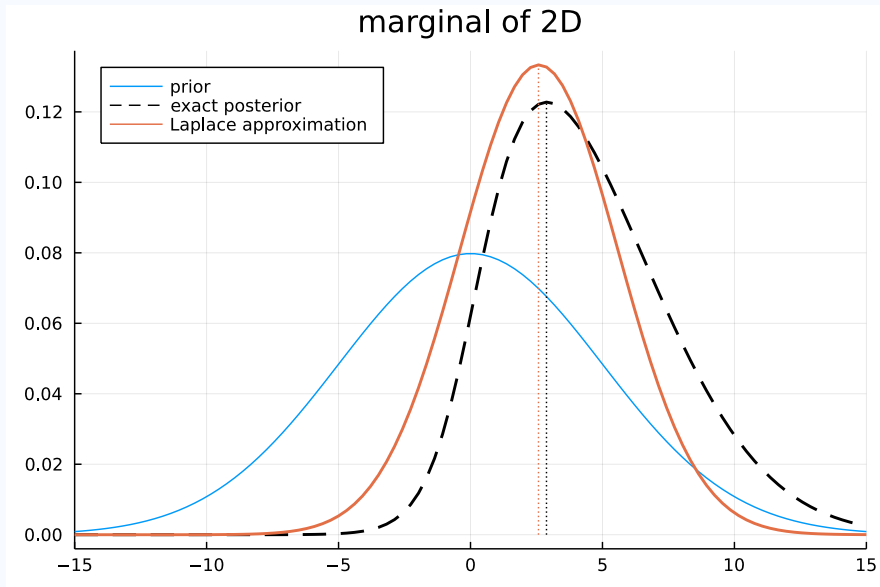
$$p(f | y) \approx \mathcal{N}(f | \hat{f}, -(\mathrm{d}^2 \log h / \mathrm{d}f^2)^{-1}) = q(f)$$



Laplace in 2D example



Laplace in 2D: marginals



Laplace approximation: important properties

- find mode: Newton's method
- match curvature (Hessian) at mode
- “point estimate++”
 - + simple, fast
 - poor approximation if mode is not representative (e.g. Bernoulli)
 - may not converge for non-log-concave likelihoods
[Hartmann and Vanhatalo, 2018]

Choosing μ and Σ for $q(\mathbf{f})$

$$p(\mathbf{f} | \mathbf{y}) \approx q(\mathbf{f}) = \mathcal{N}(\mathbf{f} | \mu = ?, \Sigma = ?)$$

locally: match mean &
variance at point

globally: minimise divergence

Laplace
approximation

Variational
Inference (VI)

Expectation
Propagation (EP)

Variational Bayes (VB)

Variational Inference (VI)

Variational inference: the big picture

Recipe for approximating intractable distribution
 $p \in \mathcal{P}$

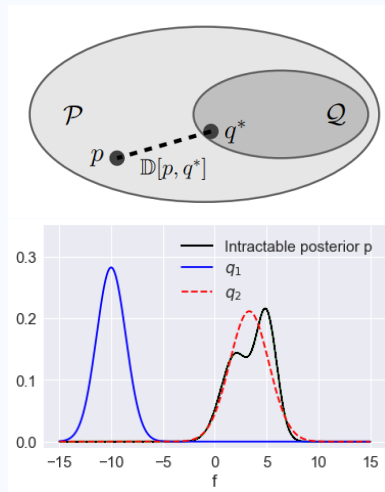
1. Define some “simple” family of distributions $\mathcal{Q}$.
2. Define some way to compute a “distance” $\mathbb{D}[p, q]$ between intractable distribution p and each distribution $q \in \mathcal{Q}$

$$\mathbb{D}[p, q_1] > \mathbb{D}[p, q_2]$$

3. Search for $q \in \mathcal{Q}$ such that $\mathbb{D}[p, q]$ is minimized

$$q^* = \arg \min_{q \in \mathcal{Q}} \mathbb{D}[p, q]$$

4. Use q^* as an approximation of p



How to “measure distances” between distributions?

Here: *Kullback–Leibler (KL) divergence*

$$\mathbb{D}[p, q] := \text{KL}[q \parallel p] = \int q(f) \log \frac{q(f)}{p(f)} \mathrm{d}f = \mathbb{E}_q \left[\log \frac{q(f)}{p(f)} \right]$$

Important properties:

1. Non-symmetric: $\text{KL}[q \parallel p] \neq \text{KL}[p \parallel q]$
2. Positive: $\text{KL} \geq 0$ (Gibbs’ inequality)
3. Minimum: $\text{KL}[q \parallel p] = 0 \iff q \equiv p$.

$$q(\mathbf{f}) = \mathcal{N}(\mu, \Sigma)$$

$$\operatorname{argmin}_{\mu, \Sigma} \operatorname{KL} [q(\mathbf{f}) \| p(\mathbf{f} | \mathbf{y})]$$

Variational inference: Minimizing $\text{KL}[q(\mathbf{f}) \| p(\mathbf{f} | \mathbf{y})]$

$$\begin{aligned}\text{KL}[q(\mathbf{f}) \| p(\mathbf{f} | \mathbf{y})] &= \int q(\mathbf{f}) \left[\log \frac{q(\mathbf{f})}{p(\mathbf{f} | \mathbf{y})} \right] d\mathbf{f} \\&= \int q(\mathbf{f}) [\log q(\mathbf{f}) - \log p(\mathbf{f} | \mathbf{y})] d\mathbf{f} \\&= \int q(\mathbf{f}) \left[\underbrace{\log q(\mathbf{f}) - \log p(\mathbf{f})}_{\text{KL}[q(\mathbf{f}) \| p(\mathbf{f})]} - \log p(\mathbf{y} | \mathbf{f}) + \log p(\mathbf{y}) \right] d\mathbf{f} \\&= \int q(\mathbf{f}) \left[\log \frac{q(\mathbf{f})}{p(\mathbf{f})} \right] d\mathbf{f} - \int q(\mathbf{f}) [\log p(\mathbf{y} | \mathbf{f})] d\mathbf{f} + \log p(\mathbf{y}) \\&= \text{KL}[q(\mathbf{f}) \| p(\mathbf{f})] - \int q(\mathbf{f}) [\log p(\mathbf{y} | \mathbf{f})] d\mathbf{f} + \log p(\mathbf{y}) \\ \log p(\mathbf{y}) &= \int q(\mathbf{f}) [\log p(\mathbf{y} | \mathbf{f})] d\mathbf{f} - \text{KL}[q(\mathbf{f}) \| p(\mathbf{f})] + \text{KL}[q(\mathbf{f}) \| p(\mathbf{f} | \mathbf{y})]\end{aligned}$$

Variational inference: Minimizing $\text{KL}[q(\mathbf{f}) \parallel p(\mathbf{f} \mid \mathbf{y})]$ by bounding

$$\begin{aligned}\log p(\mathbf{y}) &= \underbrace{\int q(\mathbf{f}) [\log p(\mathbf{y} \mid \mathbf{f})] d\mathbf{f} - \text{KL}[q(\mathbf{f}) \parallel p(\mathbf{f})]}_{\mathcal{L}[q]} + \underbrace{\text{KL}[q(\mathbf{f}) \parallel p(\mathbf{f} \mid \mathbf{y})]}_{\geq 0} \\ &\geq \int q(\mathbf{f}) [\log p(\mathbf{y} \mid \mathbf{f})] d\mathbf{f} - \text{KL}[q(\mathbf{f}) \parallel p(\mathbf{f})] = \mathcal{L}[q]\end{aligned}$$

- $\log p(\mathbf{y})$ is a constant
- $\mathcal{L}[q]$ does **not** depend on $p(\mathbf{f} \mid \mathbf{y})$
- $\mathcal{L}[q] \leq \log p(\mathbf{y})$, so $\mathcal{L}[q]$ is *lower bound* on marginal log likelihood $\log p(\mathbf{y})$
- Maximizing $\mathcal{L}[q]$ is equivalent to minimizing $\text{KL}[q(\mathbf{f}) \parallel p(\mathbf{f} \mid \mathbf{y})]$

Key take-away: we can fit variational approximation q by optimizing $\mathcal{L}$

$$\log p(\mathbf{y}) \geq \mathcal{L}[q] = \underbrace{\int q(\mathbf{f}) [\log p(\mathbf{y} | \mathbf{f})] d\mathbf{f}}_{\text{data fit}} - \underbrace{\text{KL}[q(\mathbf{f}) \parallel p(\mathbf{f})]}_{\text{regularization}}$$

$\mathcal{L}[q]$ often called the *Evidence Lower Bound* (ELBO)

- We approximate $p(\mathbf{f} | \mathbf{y}) \approx q(\mathbf{f}) = \mathcal{N}(\mathbf{f} | \mu = ?, \Sigma = ?)$
- Defining $\lambda = \{\mu, \Sigma\}$, we can write $\mathcal{L}[q] = \mathcal{L}(\lambda)$
- In practice, we optimize $\mathcal{L}(\lambda)$ using gradient-based methods

Likelihood term (data fit)

Integral separates for a factorizing likelihood:

$$\begin{aligned} & \int q(\mathbf{f}) [\log p(\mathbf{y} | \mathbf{f})] d\mathbf{f} \\ &= \sum_{n=1}^N \int q(f_n) [\log p(y_n | f_n)] df_n \end{aligned}$$

Sum over 1D integrals:

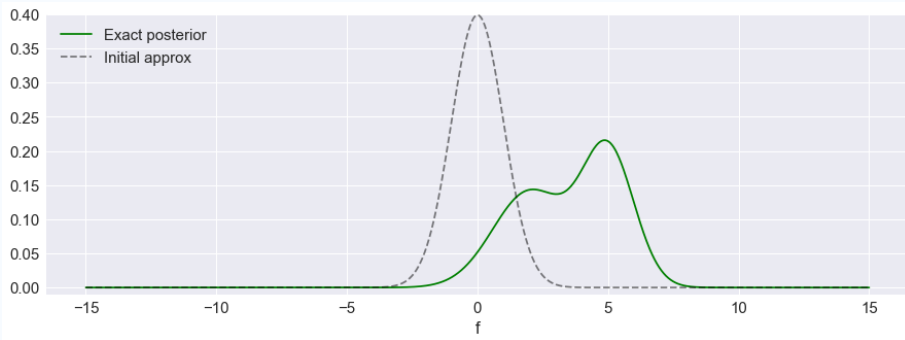
- analytic for some (e.g. Exponential, Gamma, Poisson)
- fast and accurate to approximate numerically (for example Gauss–Hermite quadrature)
- Monte Carlo (e.g. multi-class classification)

Take away #2: We can tractably optimize the bound for non-Gaussian likelihoods

1D Toy example

Assume model $p(y, f)$ with some intractable posterior $p(f | y)$

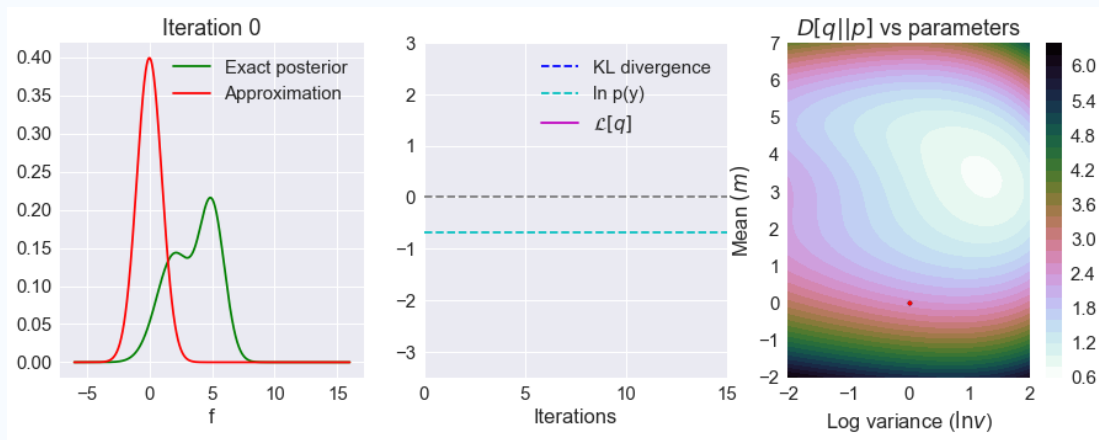
- In 1D: $\mathcal{Q}$ is the set of univariate Gaussians, i.e. $q_{\lambda}(f) = \mathcal{N}(f | m, v)$, and $\lambda = \{m, v\}$
- Initialization: $q(f) = \mathcal{N}(f | 0, 1)$



1D Toy example

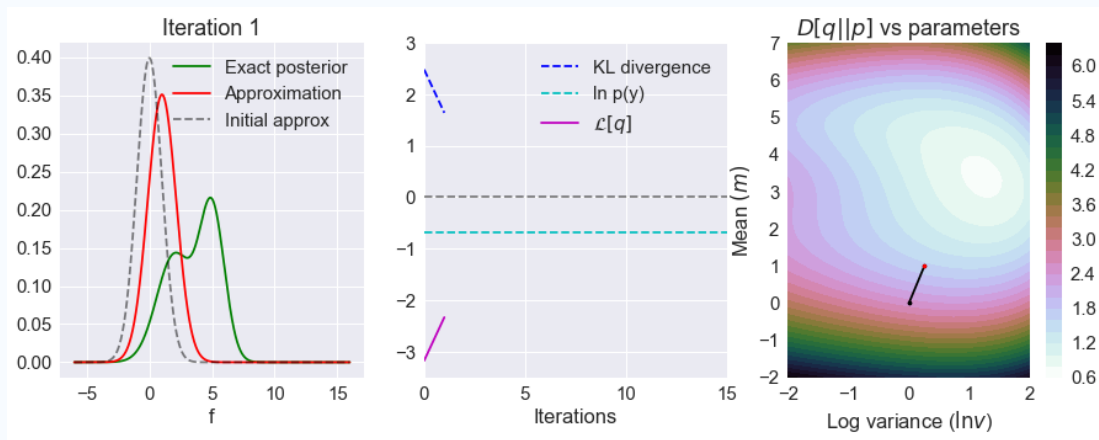
■ Gradient ascent: $\lambda_{i+1} = \lambda_i + \eta \nabla_{\lambda} \mathcal{L}(\lambda)$

■ $\log p(\mathbf{y}) = \mathcal{L}(\lambda) + \mathbb{D}[q_{\lambda}(\mathbf{f}) \parallel p(\mathbf{f} \mid \mathbf{y})] \geq \mathcal{L}(\lambda)$



1D Toy example

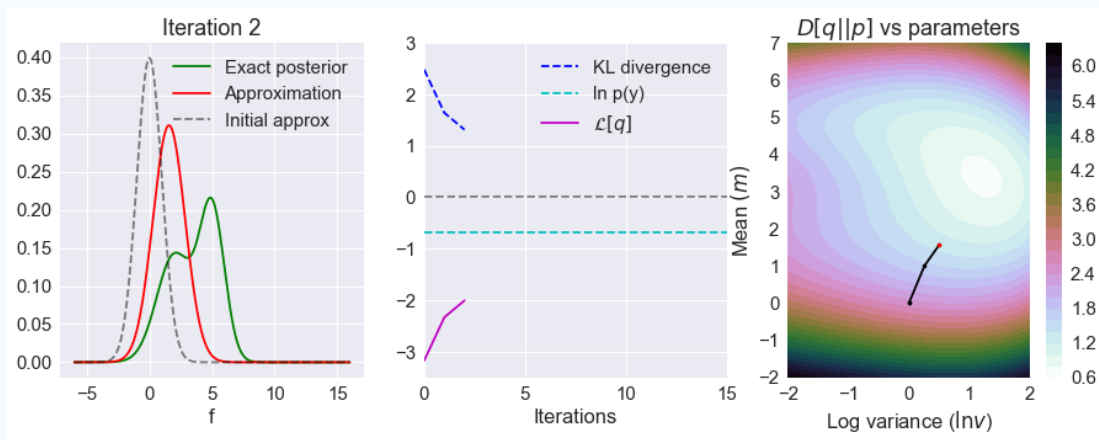
- Gradient ascent: $\lambda_{i+1} = \lambda_i + \eta \nabla_{\lambda} \mathcal{L}(\lambda)$
- $\log p(\mathbf{y}) = \mathcal{L}(\lambda) + \mathbb{D}[q_{\lambda}(\mathbf{f}) \parallel p(\mathbf{f} \mid \mathbf{y})] \geq \mathcal{L}(\lambda)$



1D Toy example

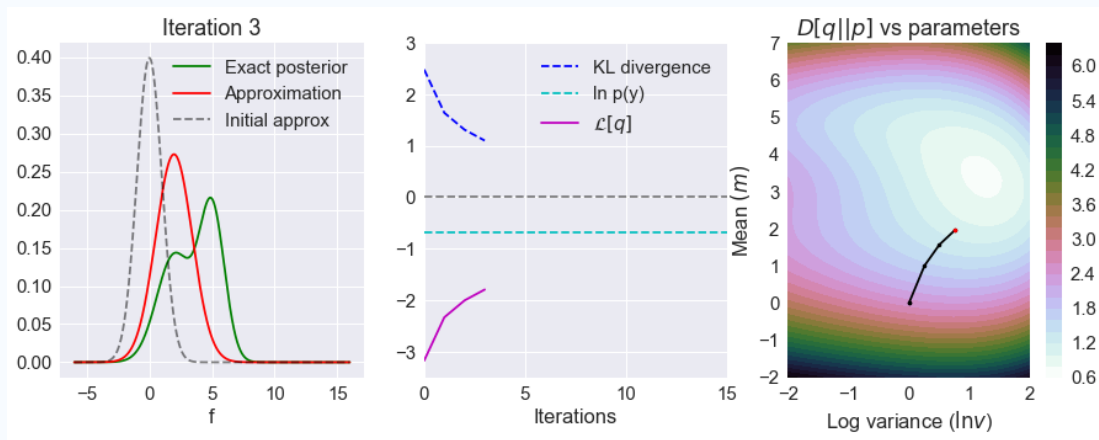
■ Gradient ascent: $\lambda_{i+1} = \lambda_i + \eta \nabla_{\lambda} \mathcal{L}(\lambda)$

■ $\log p(\mathbf{y}) = \mathcal{L}(\lambda) + \mathbb{D}[q_{\lambda}(\mathbf{f}) \parallel p(\mathbf{f} \mid \mathbf{y})] \geq \mathcal{L}(\lambda)$



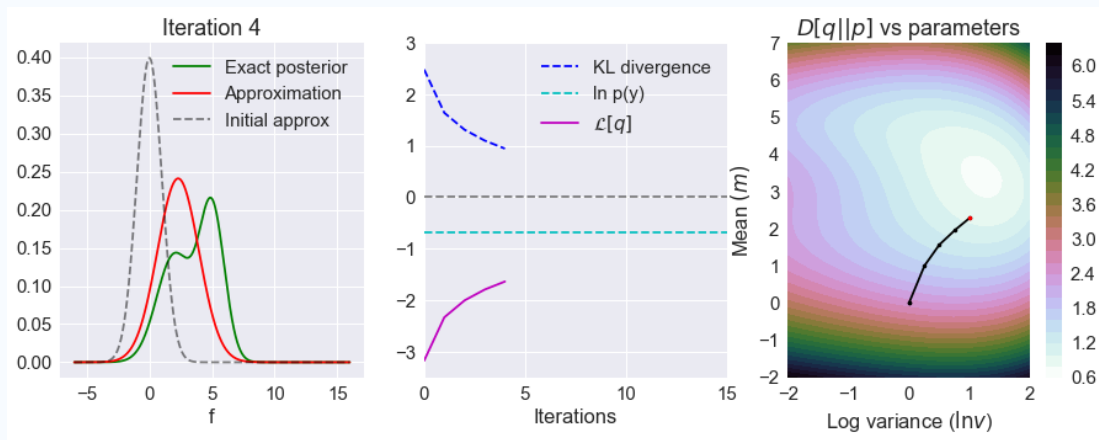
1D Toy example

- Gradient ascent: $\lambda_{i+1} = \lambda_i + \eta \nabla_{\lambda} \mathcal{L}(\lambda)$
- $\log p(\mathbf{y}) = \mathcal{L}(\lambda) + \mathbb{D}[q_{\lambda}(\mathbf{f}) \parallel p(\mathbf{f} \mid \mathbf{y})] \geq \mathcal{L}(\lambda)$



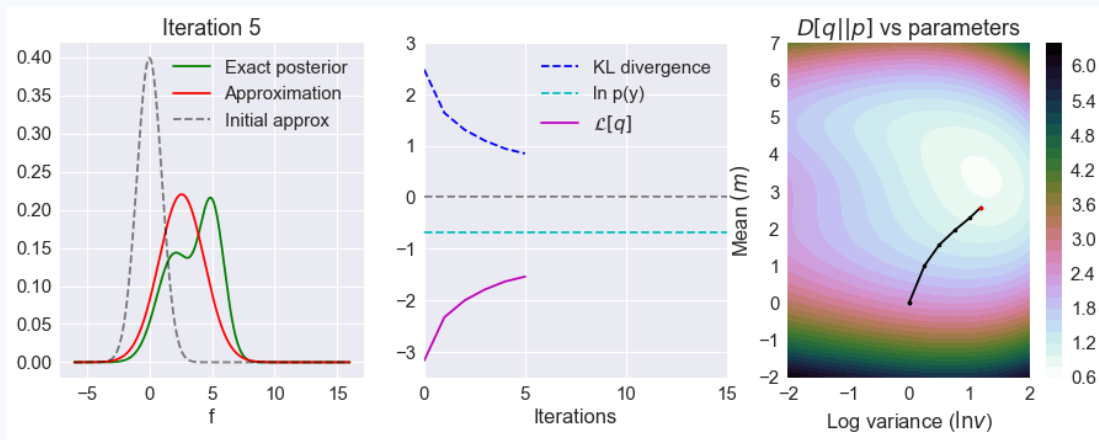
1D Toy example

- Gradient ascent: $\lambda_{i+1} = \lambda_i + \eta \nabla_{\lambda} \mathcal{L}(\lambda)$
- $\log p(\mathbf{y}) = \mathcal{L}(\lambda) + \mathbb{D}[q_{\lambda}(\mathbf{f}) \parallel p(\mathbf{f} \mid \mathbf{y})] \geq \mathcal{L}(\lambda)$



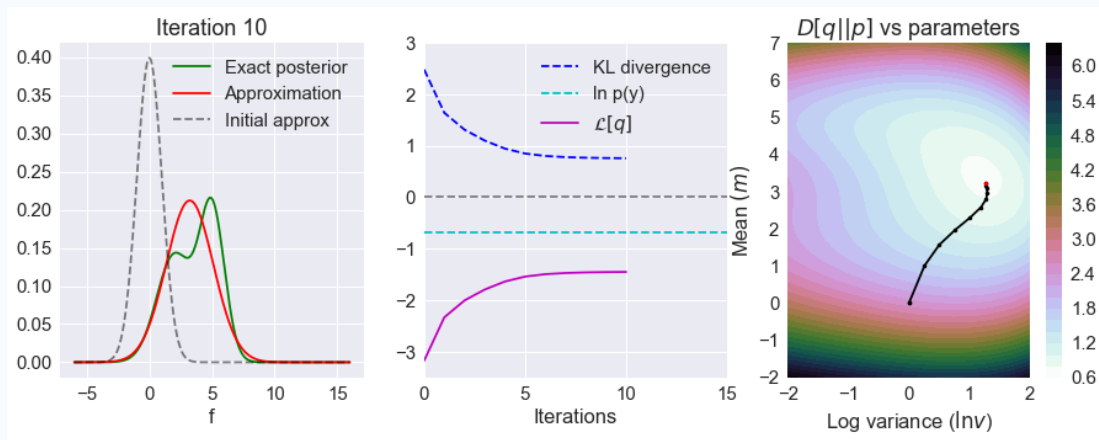
1D Toy example

- Gradient ascent: $\lambda_{i+1} = \lambda_i + \eta \nabla_{\lambda} \mathcal{L}(\lambda)$
- $\log p(\mathbf{y}) = \mathcal{L}(\lambda) + \mathbb{D}[q_{\lambda}(\mathbf{f}) \parallel p(\mathbf{f} \mid \mathbf{y})] \geq \mathcal{L}(\lambda)$



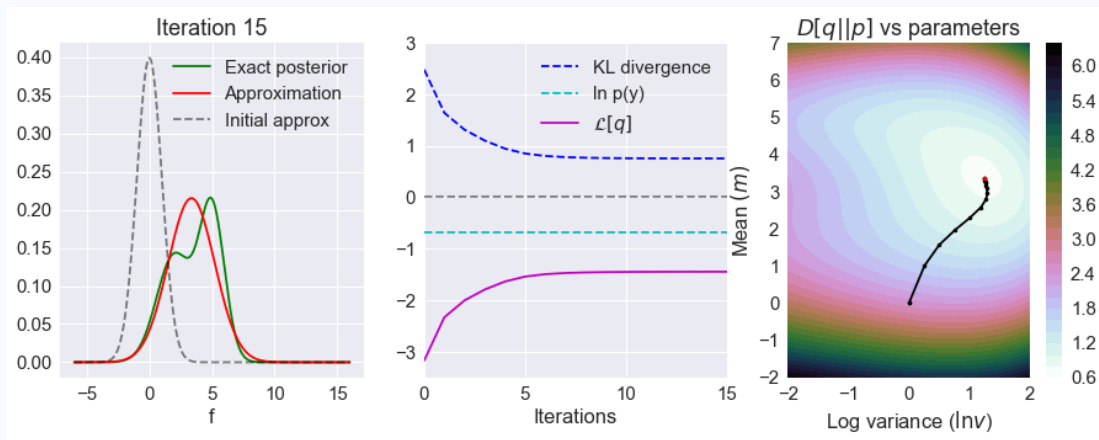
1D Toy example

- Gradient ascent: $\lambda_{i+1} = \lambda_i + \eta \nabla_{\lambda} \mathcal{L}(\lambda)$
- $\log p(\mathbf{y}) = \mathcal{L}(\lambda) + \mathbb{D}[q_{\lambda}(\mathbf{f}) \parallel p(\mathbf{f} \mid \mathbf{y})] \geq \mathcal{L}(\lambda)$



1D Toy example

- Gradient ascent: $\lambda_{i+1} = \lambda_i + \eta \nabla_{\lambda} \mathcal{L}(\lambda)$
- $\log p(\mathbf{y}) = \mathcal{L}(\lambda) + \mathbb{D}[q_{\lambda}(\mathbf{f}) \parallel p(\mathbf{f} \mid \mathbf{y})] \geq \mathcal{L}(\lambda)$



Variational inference: important properties

- principled: directly minimising divergence from true posterior
- mode-seeking (e.g. multi-modal posterior: fits just one, if q is unimodal)
 - + minimises a true lower bound $\rightarrow$ convergence
 - underestimates variance

$$p(\mathbf{f} | \mathbf{y}) \approx q(\mathbf{f}) = \mathcal{N}(\mathbf{f} | \mu = ?, \Sigma = ?)$$

- ✓ $\min \text{KL}[q(\mathbf{f}) \| p(\mathbf{f} | \mathbf{y})]$: Variational Inference
- 2. $\min \text{KL}[p(\mathbf{f} | \mathbf{y}) \| q(\mathbf{f})]$: **Expectation Propagation**

Expectation Propagation (EP)

Expectation Propagation

Can we minimise KL divergence in opposite direction?

$$q(\mathbf{f}) = \operatorname{argmin}_{\mu, \Sigma} \operatorname{KL} [p(\mathbf{f} | \mathbf{y}) \| q(\mathbf{f})] = \operatorname{argmin}_{\mu, \Sigma} \int p(\mathbf{f} | \mathbf{y}) \left[\log \frac{p(\mathbf{f} | \mathbf{y})}{q(\mathbf{f})} \right] d\mathbf{f}$$

Exact posterior: $p(\mathbf{f} | \mathbf{y}) \propto p(\mathbf{f}) \prod_{n=1}^N p(y_n | f_n)$

Approximate posterior: $q(\mathbf{f}) \propto p(\mathbf{f}) \prod_{n=1}^N t_n(f_n)$

$$t_n = Z_n \mathcal{N}(f_n | \tilde{\mu}_n, \tilde{\sigma}_n^2) \quad \text{“sites”}$$

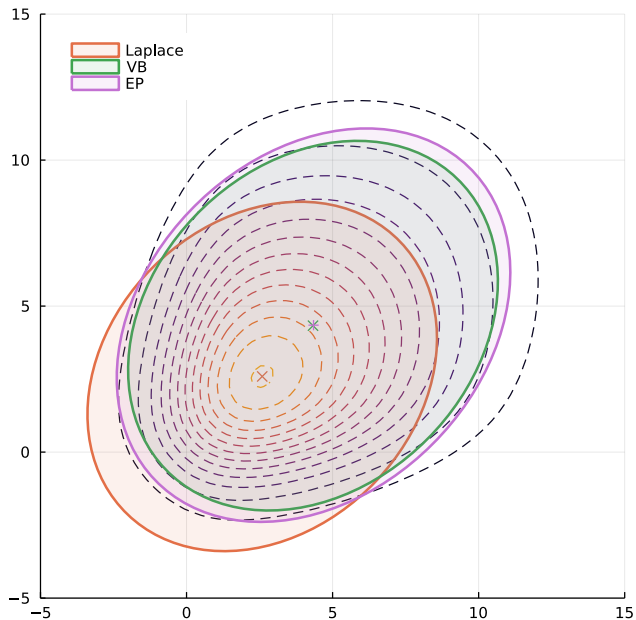
- Expectation propagation iteratively updates the sites for each data point
- minimizes KL of local approximations to the posterior:

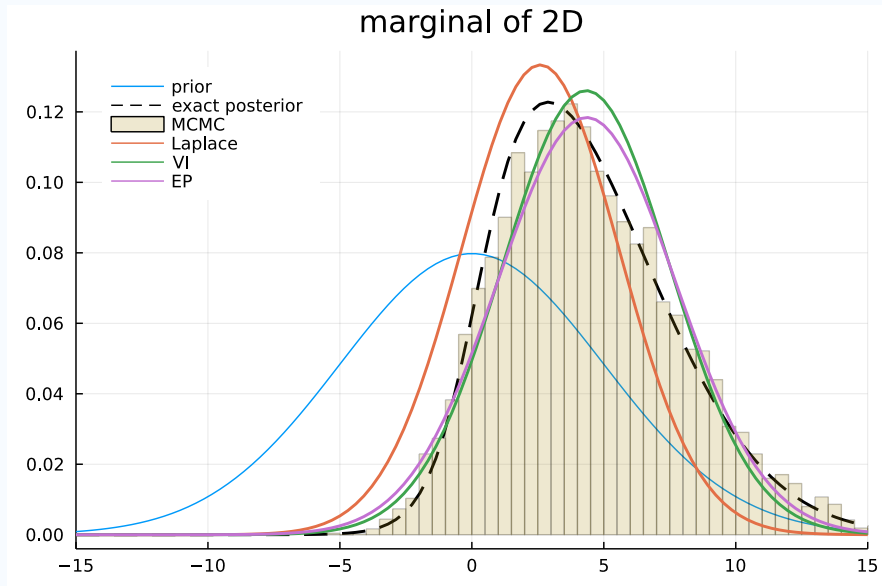
$$\min \operatorname{KL} [q(f_n) \frac{p(y_n | f_n)}{t_n(f_n)} \| q(f_n) \frac{t'_n(f_n)}{t_n(f_n)}]$$

Expectation Propagation: important properties

- multiple passes required to converge
- moment-matching (e.g. covering multiple modes)
 - + effective for classification
 - not guaranteed to converge
 - updates may be invalid (non-log-concave likelihoods) [Jylänki et al., 2011]

Comparison 2D





- ✓ Gaussian processes with Gaussian likelihood
- ✓ What is the likelihood? Connecting observations and Gaussian process prior
- ✓ Non-Gaussian likelihoods: what happens to the posterior?
- ✓ How to approximate the intractable
 - ✓ with samples: MCMC
 - ✓ with Gaussians
 - Laplace
 - Variational Inference
 - Expectation Propagation

5. Comparison

Comparison

Comparison

MCMC

- ▶ samples
- ▶ gold standard
- ▶ slow

Laplace

- ▶ $\mathcal{N}$ = curvature at mode
- ▶ simple & fast
- ▶ often poor approximation

Variational Inference

- ▶ $\mathcal{N}$ minimises $\text{KL}[q(\mathbf{f})||p(\mathbf{f} | \mathbf{y})]$
- ▶ principled, any likelihood
- ▶ underestimates variance

Expectation Propagation

- ▶ $\mathcal{N}$ matches marginal moments
- ▶ good calibration in classification
- ▶ may not converge

What about hyperparameters

(kernel: lengthscales, function scale, etc.; likelihood: noise scale, ...)?

- **MCMC:** priors on hyperparameters, integrate out everything
- **Gaussian approximations:** approximations to marginal likelihood
 - ▶ (may be biased)

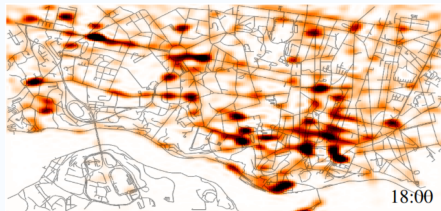
What we did not cover...

- More complex likelihoods (heteroskedastic, zero-inflated, multi-stage...)
- Marginal likelihood approximations for hyperparameter learning [Nickisch and Rasmussen, 2008, Li et al., 2023]
- How parametrisation affects Gaussianity of $p(\mathbf{f} | \mathbf{y})$
- Connections between EP and VI (“PowerEP”, CVI dual parameterization) [Bui et al., 2017, Adam et al., 2021]
- Other divergences, generalised VI, ...
- Combinations of MCMC and variational methods
- Augmenting likelihood with auxiliary variable
→ conditionally conjugate model [Galy-Fajou et al., 2020]

Take-aways

We can...

- create **richer models** with likelihoods beyond the Gaussian
- **learn latent functions** that form the connection between data points
- handle the non-Gaussian posterior with **approximations**
- **trade off** speed, accuracy, and ease-of-use



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